

TRANSPONDER LANDING SYSTEM (TLS) INSTALLATION MANUAL



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NOTICE

Advanced Navigation and Positioning Corporation has attempted to describe in this manual all areas of possible danger to personnel in connection with the use of this equipment. This document refers to government and professional technical standards: Consult standards prevailing in the region of installation to assure compliance with local requirements.

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**ENGINEERING CHANGE ORDER RECORD
TRANSPONDER LANDING SYSTEM (TLS) INSTALLATION MANUAL**

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Revision	Issue Date	Revision Description	ECN #
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SECTION 1: INTRODUCTION

1.1 Document Overview

This document contains information needed to the install and configure the Transponder Landing System (TLS) and Transportable TLS.

1.2 ANPC TLS Manuals

The ANPC TLS is accompanied by four primary manuals:

- Document # 020-00071, TLS Calibration Manual
- Document # 020-00072, TLS Operators Manual
- Document # 020-00073, TLS Installation Manual
- Document # 020-00074, TLS Maintenance Manual

The ANPC TTLS includes two additional manuals:

- Document 020-00076 Container System Deployment
- Document 020-00077 TTLS Packing Instructions

SECTION 2: INSTALLATION

Where requested, record verification in the Test Results Register (TRR).

In addition to the general installation instructions that follow, technicians are required to use assembly drawings for details necessary to correctly perform the installation of the TLS.

2.1 TLS Siting Criteria

This section contains a summary of the ANPC siting criteria document 300-00036. Placement of equipment beyond the limits shown below may be acceptable after consultation with ANPC and the siting criteria document.

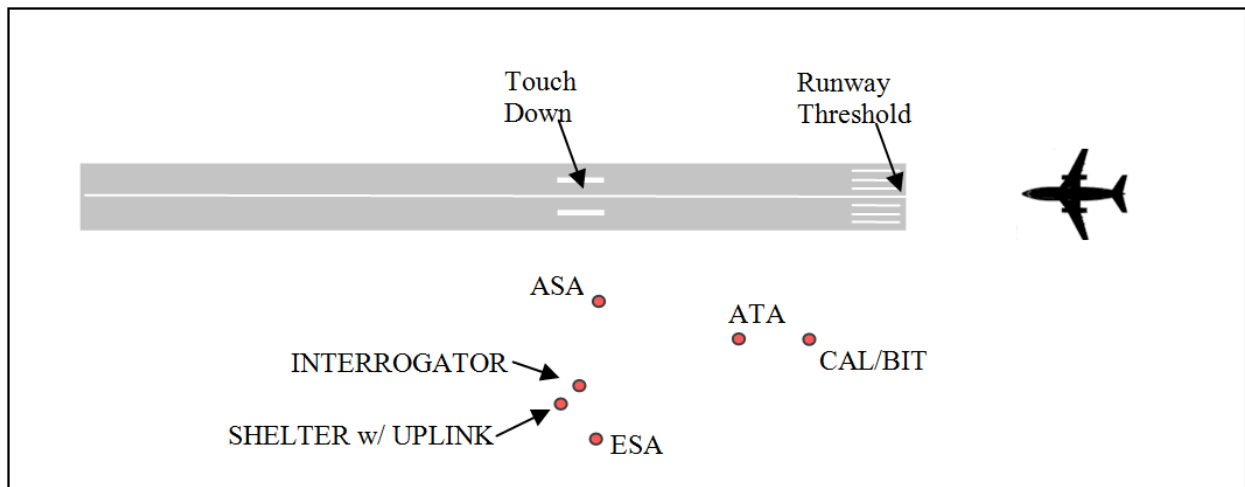


Figure 2-1 Standard positions for TLS components on one side of a runway

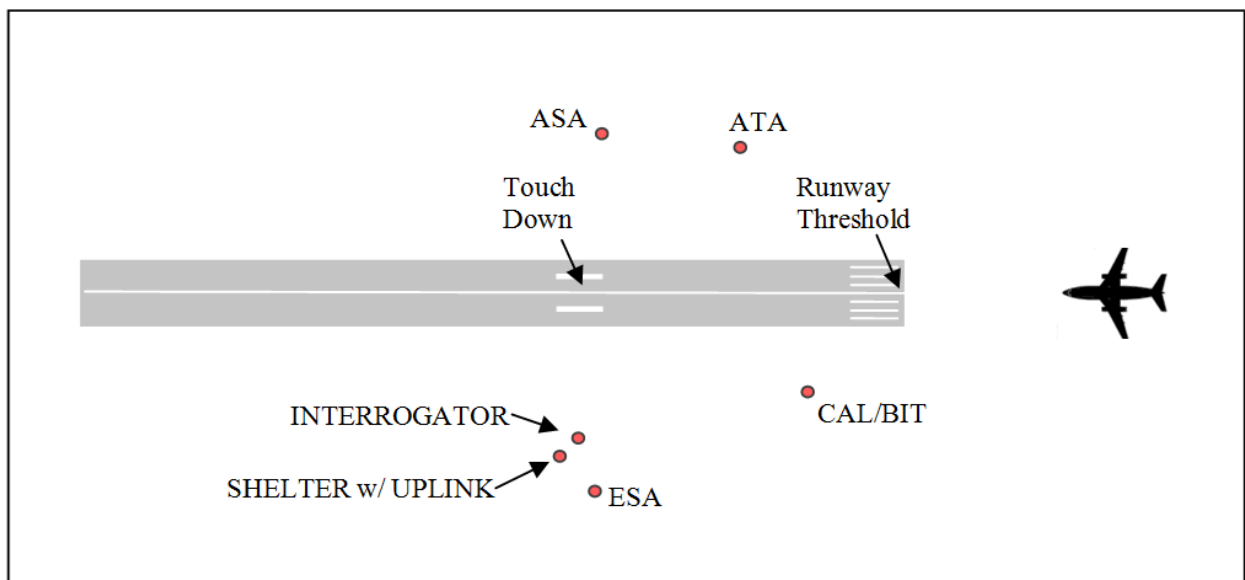


Figure 2-2 Standard positions for TLS components on two sides of a runway

Table 2-1 Summary of key siting parameters

	Minimum	Standard	Maximum
ASA w.r.t Touch Down (TD)	TD-100m	TD-50m	TD+75m
ASA to ESA separation	50 m (option 30 m)	100 m	No spec
ESA to Cal/BIT separation	150 m	150 m	No spec
ASA to ATA	50 m	100	150 m
ESA to ATA	50 m	100	150 m
ATA to Cal/BIT	20 m	60 m	No spec
Interrogator to shelter	11 m	12 m	13m
ASA to ESA angle from perpendicular to centerline	0 degrees	5 degrees	10 degrees
ASA to centerline	50m or as OFZ permits	60 m	70m
ESA to centerline	As permitted by local regs	90 m	200 m

For any parameter noted as “no spec” the practical considerations of property boundaries and cable lengths will impose a maximum limit.

2.2 Examples of Common Siting for TLS Components



Figure 2-3 All TLS equipment on one side of runway

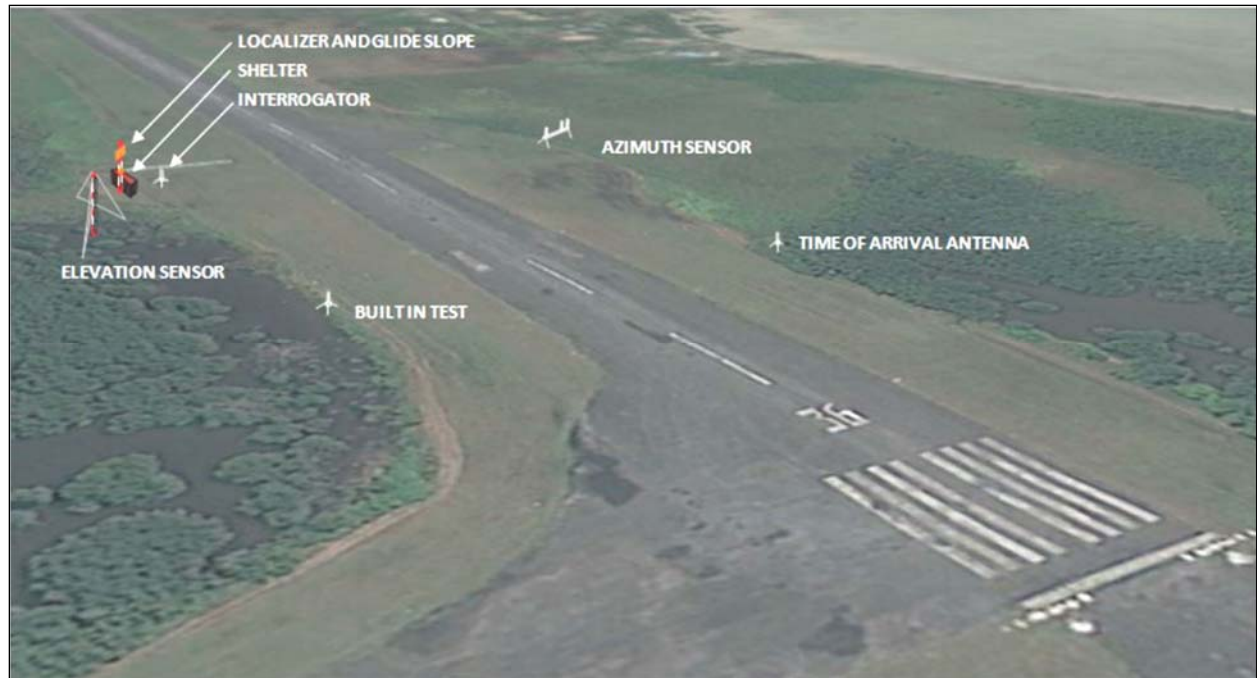


Figure 2-4 TLS equipment split on both sides of the runway

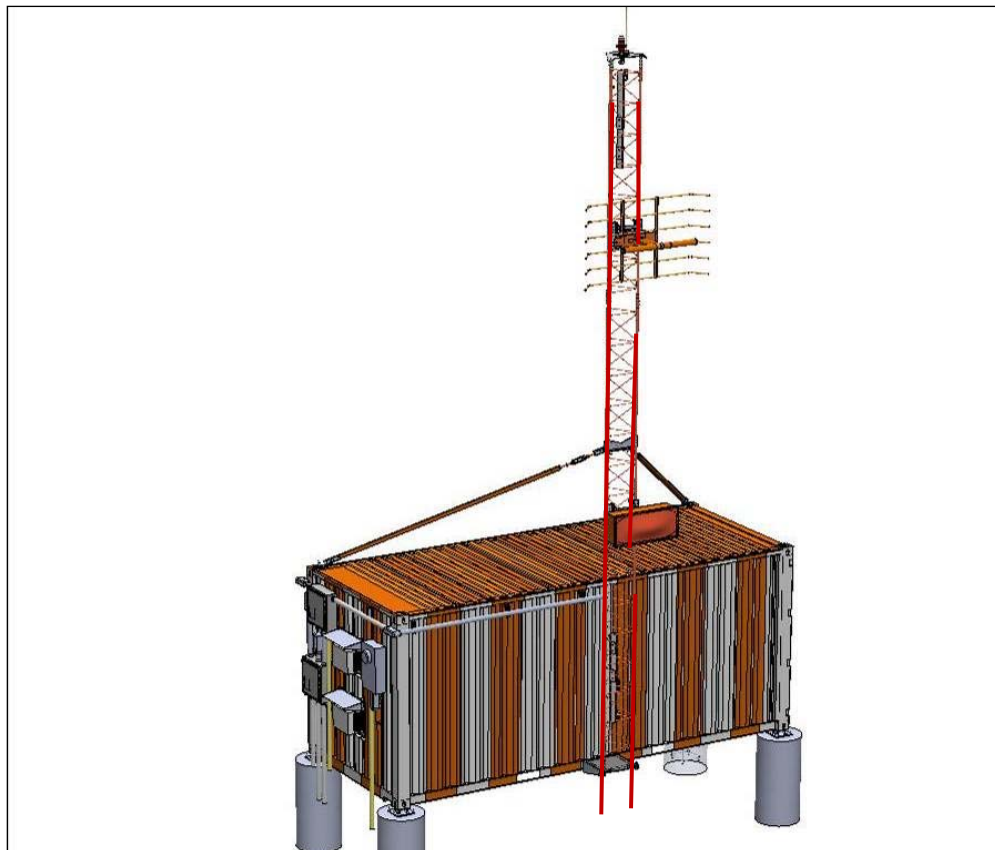


Figure 2-5 TLS shelter with hinged uplink antenna

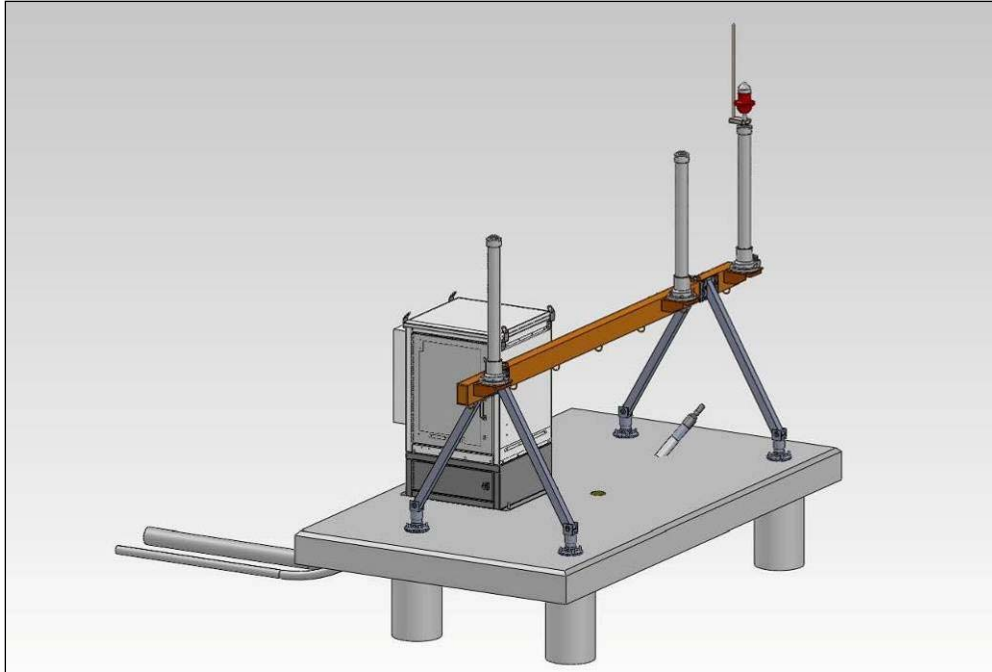


Figure 2-6 ASA with omni directional antennas

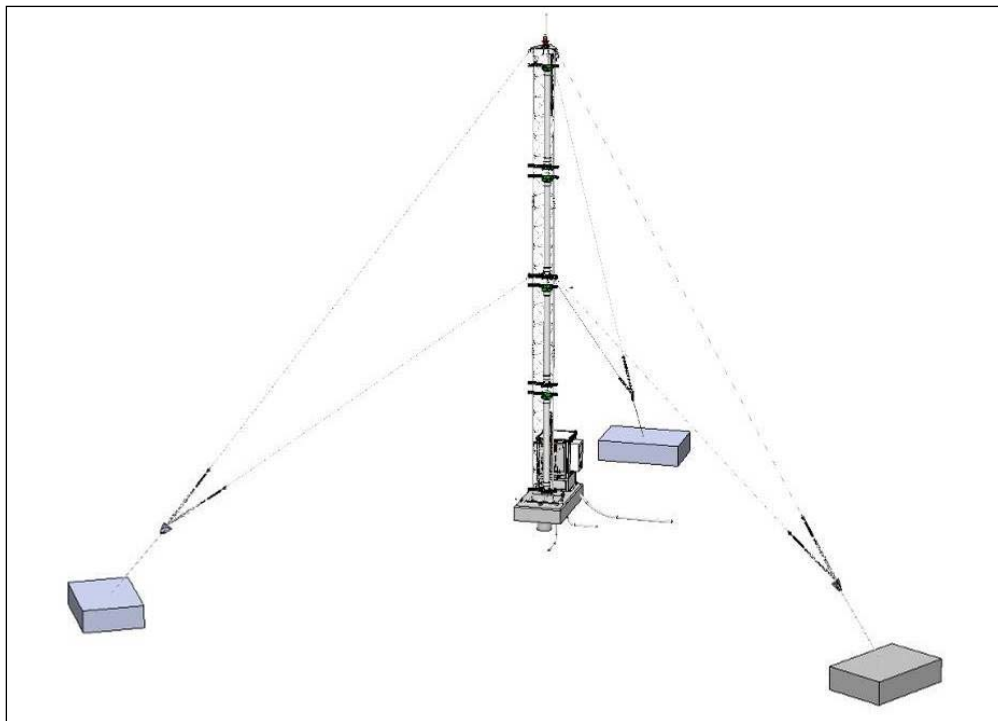


Figure 2-7 ESA with guy wire option

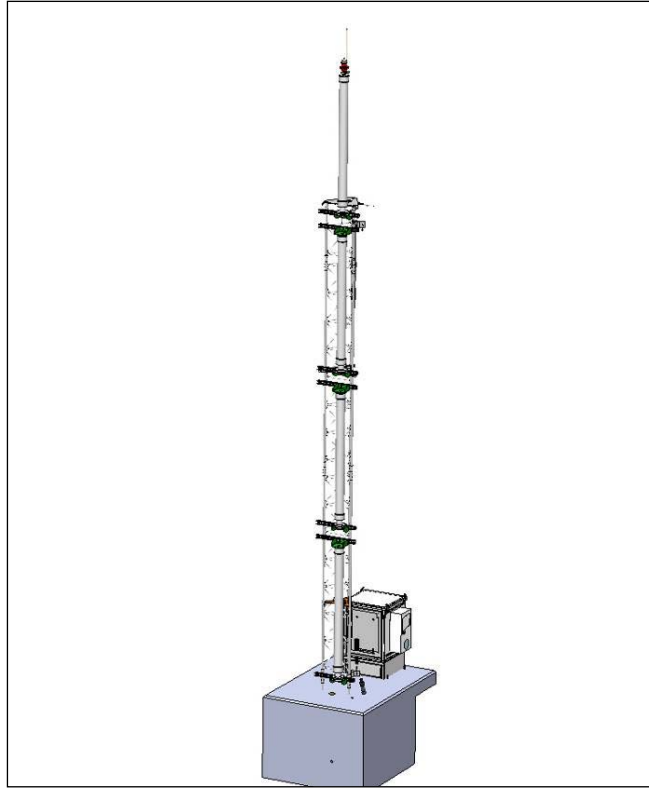


Figure 2-8 ESA option without guy wires

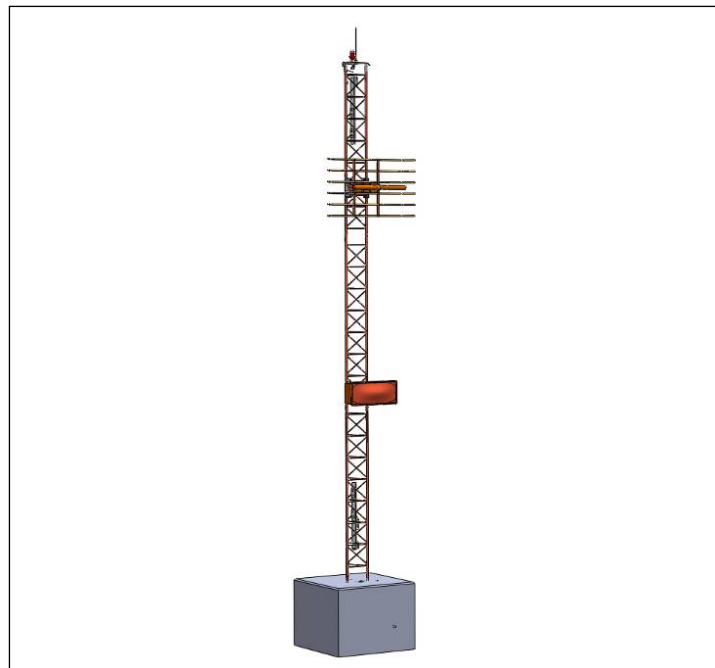


Figure 2-9 Uplink option without guy wires and without shelter attached

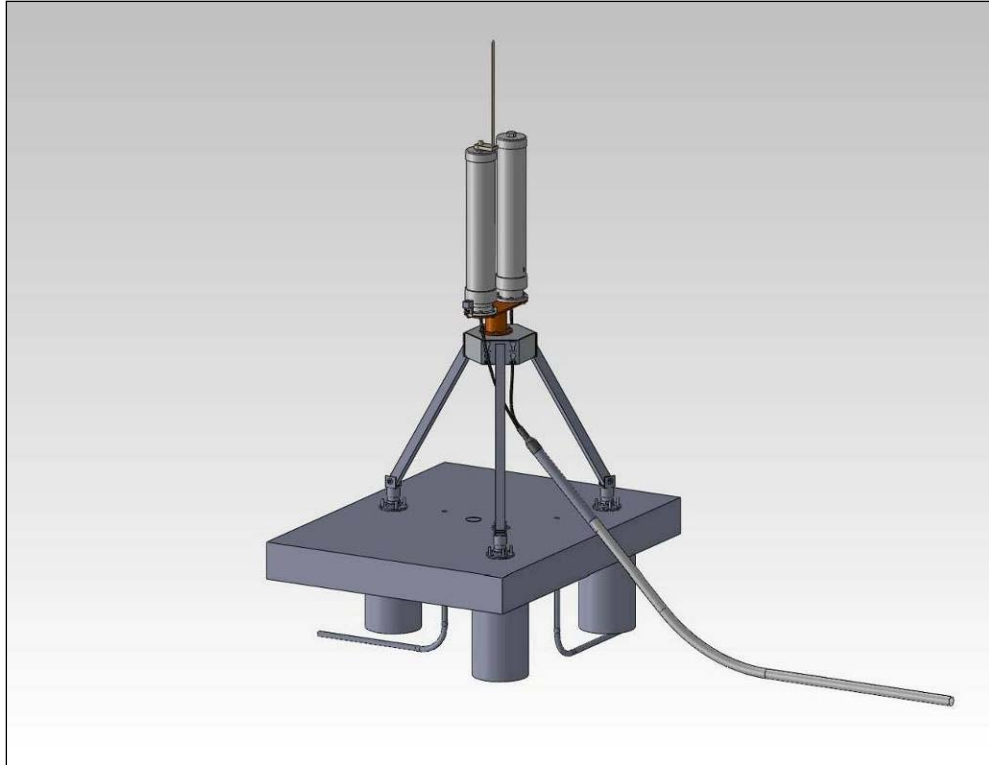


Figure 2-10 Interrogation with suppression option

NOTE: Interrogation must point towards the approach and ILS service volume; the suppression antenna beam must always point away from the approach direction.

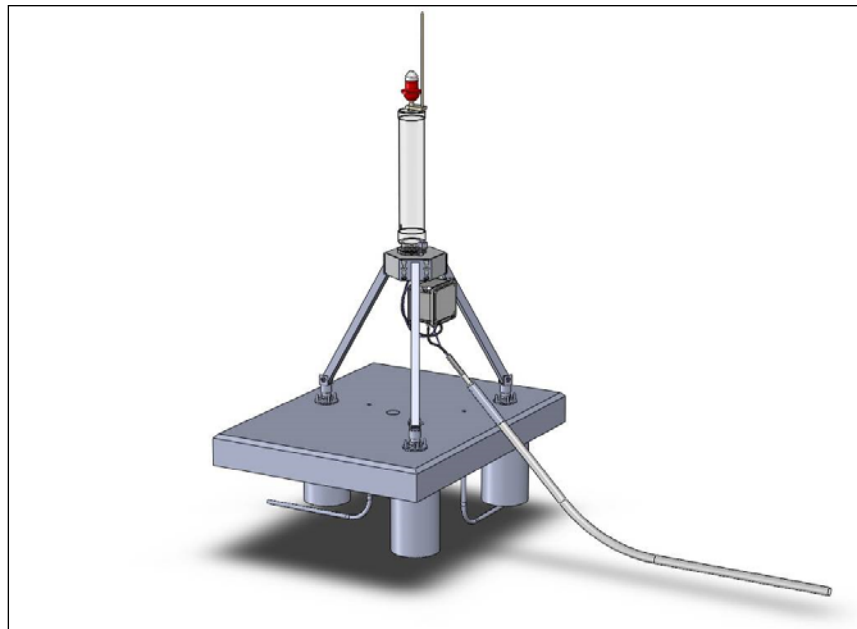


Figure 2-11 Cal/BIT with directional antenna, the antenna beam must point toward TLS

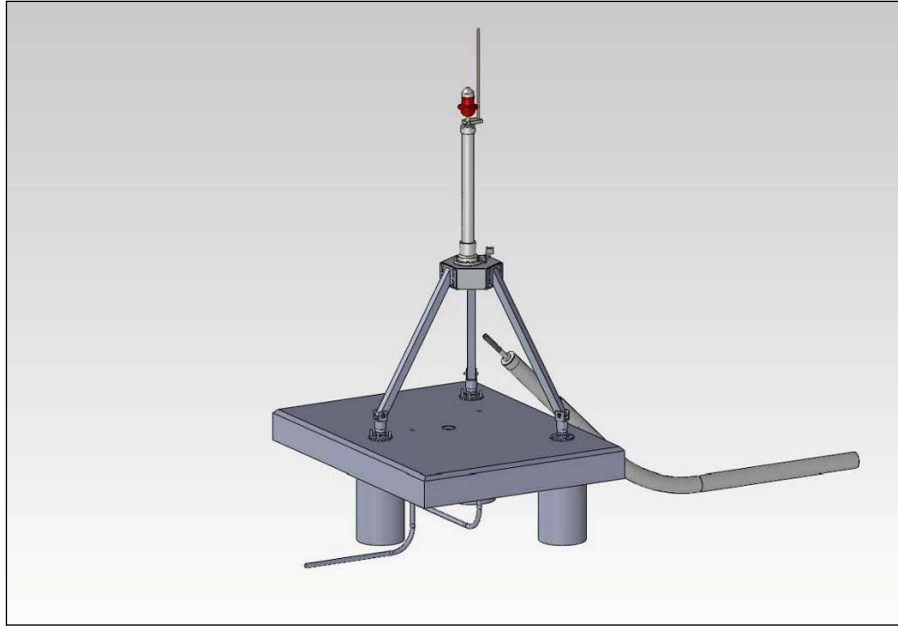


Figure 2-12 ATA with omni directional antenna

2.2.1 Equipment Locations

Due to the tower height, the ESA sensor location and its critical area should be considered first, followed by the ASA sensor and critical area, the ATA structure, the Cal/BIT structure, and then the Shelter. Final placement of components is the result of an iterative process that takes into consideration the placement requirements in Table 2-1. The critical area should also be considered early in the siting process in order to place the critical area in a location where it has minimal effect on operations. Consider the following important points when siting the TLS antennas:

- The TLS Shelter is the only point where utility power must be provided
- The critical area should be defined to have minimal impact on airport operations.
- The tall antennas of the ESA and Uplink should be outside of the Runway Safety Area and Obstacle Free Zone; other antennas are frangible and may be placed in these areas as allowed by ICAO and FAA because they are “fixed-by-function”
- For narrow properties the towers can be sited within the RSA and OFZ with waivers if no other options are available.
- Attempt to site the antennas to have minimal trenching, especially under runways; make use of existing conduits. Minimize trenching distance or complexity to bury the fiber-optic cables between the sensors and the TLS Shelter.

2.2.2 Shelter Placement

The Shelter location is typically between the ESA and ASA sensors. The shelter can be installed at any convenient location on or near the airport that satisfies the following conditions:

- The localizer, glide slope, and interrogation antennas must provide line of sight coverage to the required TLS service volume.
- Minimize trenching distance and complexity to bury the fiber-optic and power cables between the TLS shelter, the sensors and Cal/BIT.

2.2.3 TLS Component Siting

The TLS construction drawing package should be updated to show the TLS component siting. The drawing package should then be released to the contractor. The final as-built drawing package should be archived as a final document of the installation.

2.2.3.1 Concrete Foundations

All concrete foundations for TLS installation shall be completed in accordance with the site construction drawing package, unless modifications are needed to add stability in consideration of local soil conditions.

2.2.3.2 Power Requirements

The TLS electronics receive conditioned and protected AC power from the shelter UPS. Total power draw by option is listed in the figure below.

During the site design, conductor size is determined such that the voltage drop between the shelter and each component is less than 10% of the voltage.

Utility service varies according to options installed. Options include dual rack or single, split unit A/C configuration or standard.

Table 2-2 Electrical Requirements

Options	KVA		Total KVA per option	VOLTS	AMPS SERVICE	AMPS for TLS total rack(s) plus ECU's
Dual rack	Configuration	Power draw				
	UPS rack 1	2.9				
	UPS Rack 2	2.9				
	UPS Network	2.9	8.7	240 / 120	36.3 / 72.5	
Extreme hot	Split A/C	6.0	12.0	240 / NA	50 / NA	86.3 / 123
Standard A/C		0.54	0.54	NA / 120	NA / 4.5	40.8
Shelter heat	Standard	1.0	1.0	240 / 120	4.2 / 8..3	49
Deicing	Antennas	13.0	13.0	240 / NA	54 / NA	103
Single rack						
	UPS rack 1	2.9	2.9			
Extreme hot	Split A/C	6.0	12.0	240 / NA	50 / NA	52.9
Standard	a/c	0.54	0.54	NA / 120	4.5	7.4
Shelter heat	Standard	1.0	1.0	240 / 120	4.2 / 8..3	11.6
Deicing	Antennas	13.0	13.0	240 / NA	54 / NA	65.6

2.2.3.3 Installation Verification

For fixed TLS (not Transportable TLS) record the following verifications:

- System components are installed at locations per the construction drawing package.
- Concrete foundations have been constructed per the site construction drawing package.
- As-built redlines, where necessary, have been documented.

SECTION 3: INSTALLATION

3.1 Documentation Requirements

Record verification where indicated on TRR document 300-00048 Test Results Register.

In addition to the general installation instructions that follow, the installers are required to use site- specific installation drawings for details necessary to correctly perform the installation of the TLS.

3.2 Grounding

Exothermic-welds are used for the fixed TLS, lug connections using a variety of split bolts with hex head and other connections are used for rapidly deploying the Transportable TLS.

3.3 Conduit End Seals

For a fixed TLS (non-transportable) seal all open conduit ends to stop unwanted foreign material, rodents, water, etc. from entering the conduits during and after installation.

3.4 Shelter Installation

Installation of the shelter requires coordination prior to the arrival of the shelter. For a fixed TLS, the pad must be ready for the shelter and the correct hardware must be available for securing the shelter to the pad. Arrangements must be made to have the appropriate lifting equipment and operator available for when the shelter arrives on site.

3.5 Guidance Uplink Tower Installation

3.5.1 Tower Assembly

1. Attach the bottom mount plate to the shelter using the supplied hardware.
2. Assemble the three tower sections using the supplied hardware.
3. Attach the top plate to the tower using the supplied hardware.
4. Attach the assembled tower to the bottom mount plate using the supplied hardware.



Figure 3-1 Uplink tower with hinge attached to shelter

5. Attach the obstruction light and air terminal to the grounding plate and the top mounting plate.



Figure 3-2 Obstruction Light and Air Terminal

6. Attach the 32S down-conductor cable with an exothermic-welded LA lug to the obstruction light mounting plate. Run and tie the ground cable down the tower leg. A conductive bond is required on every section, and a secure connection to the tower every 3 feet.
7. Wiring for power to the obstruction light shall be secured to the tower.
8. Use the winch to stand the tower upright with the flat face of the tower and bottom mounting plate facing the runway approach.

WARNING

Use the climbing safety system when installing antennas, obstruction lights and air terminal on the uplink tower. Always use the two clip method to ensure constant attachment to the tower.

9. There are two options for securing the uplink tower
 - a. Option 1 – Installation with spars: Attach the ball hitch adaptors to the corner castings on the top corners of the shelter. Then attach the spars to the corner castings and rest the spars near the middle of the side where the uplink tower will be attached. Using the toolkit hoist or at least two helpers, rotate the uplink tower into position while a third person attaches the spars. Adjust the turnbuckles to plumb the tower.



Figure 3-3 Attaching the uplink tower with spars

- b. Option 2: Installation with uplink tower bracket: With the tower sections assembled and horizontal on the ground, attach the c-channel bracket near the expected position corresponding to the height of the shelter. Using the toolkit hoist or at least two helpers, rotate the uplink tower into position while a third person fastens the c-channel to the shelter weldments.

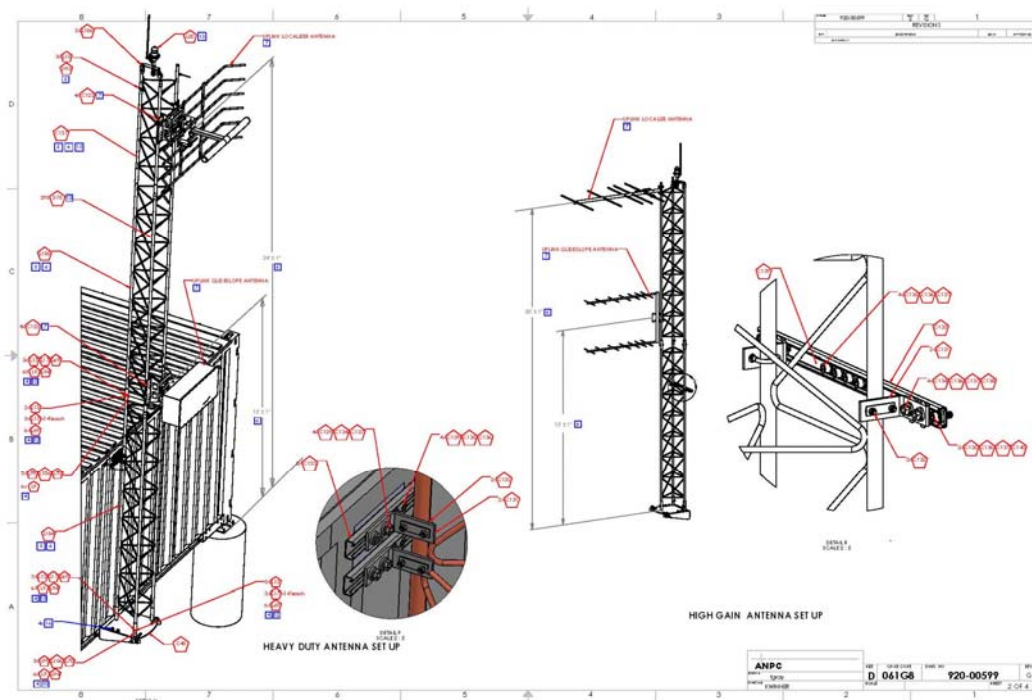


Figure 3-4 -attaching the uplink tower with a bracket to top of shelter

10. For TLS, exothermically weld the bottom end of the 32S down-conductor cable to the designated 4/0 grounding stub that comes from the Ground Electrode System.
11. Power to the obstruction light shall be connected the same day the tower is raised.

3.5.2 Glide Slope Guidance Antenna Installation

There are two antenna options, the normal gain antennas and the high gain antennas.

3.5.3 Normal gain glide slope transmit antenna installation:

1. Assemble the antenna
2. Attach the antenna to the tower with 4 U-bolts, 12 feet from the ground.
3. Ensure proper antenna alignment such that it is within $\pm 5^\circ$ of approach path.
4. Attach the coaxial cables for transmit and monitor to the antenna.
5. Route the cables down the tower and secure to the tower.

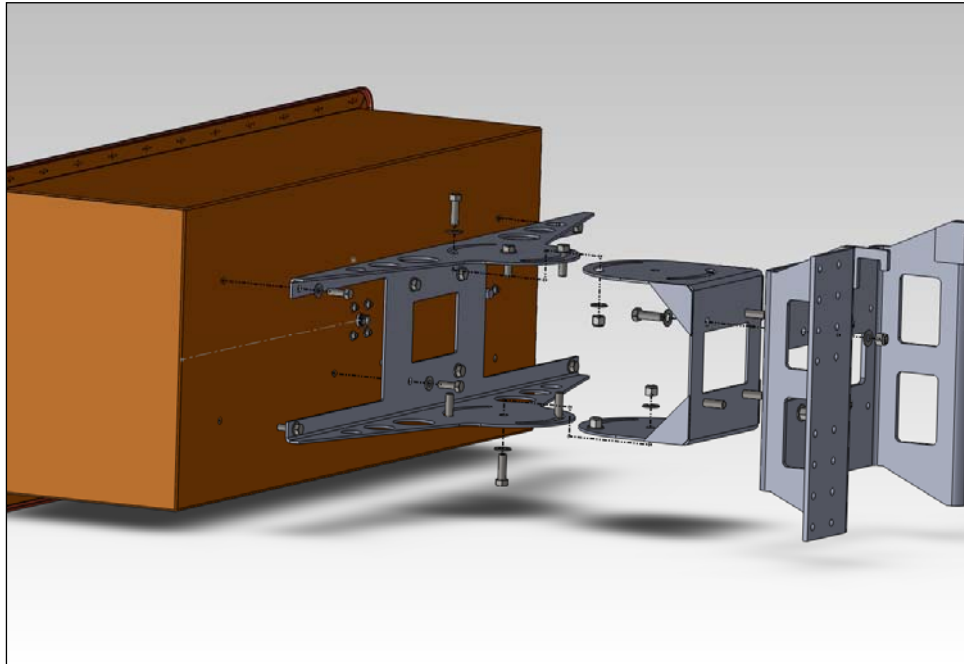


Figure 3-5 Glide slope antenna

3.5.4 Normal gain Localizer Transmit Antenna Installation

1. Assemble the antenna [Figures 3-5 to 3-7].

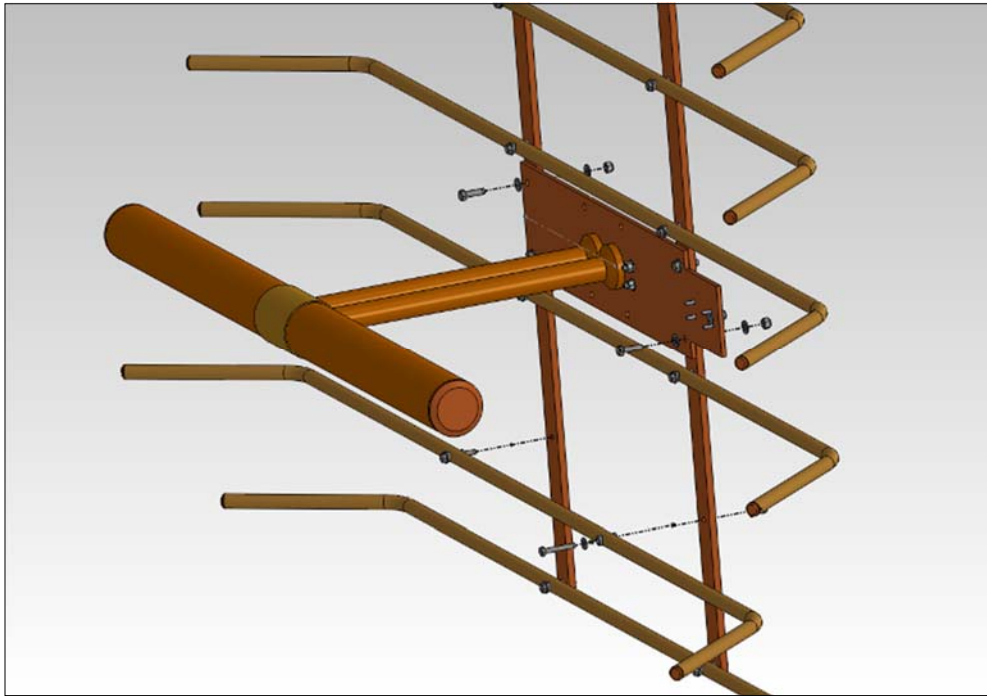


Figure 3-6 Localizer frame assembly

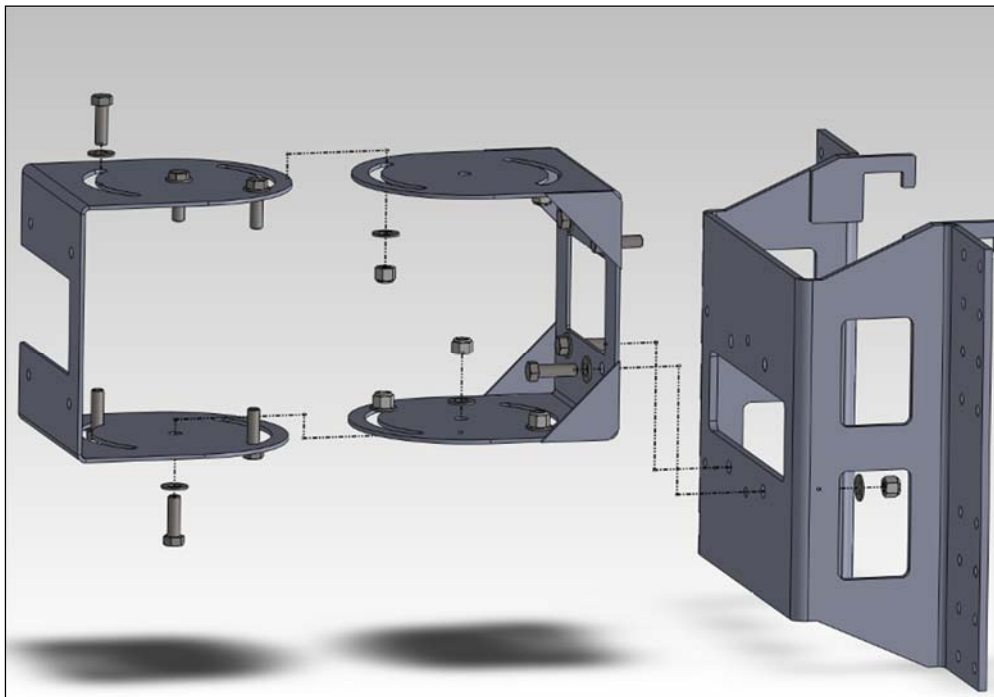


Figure 3-7 Bracket assembly

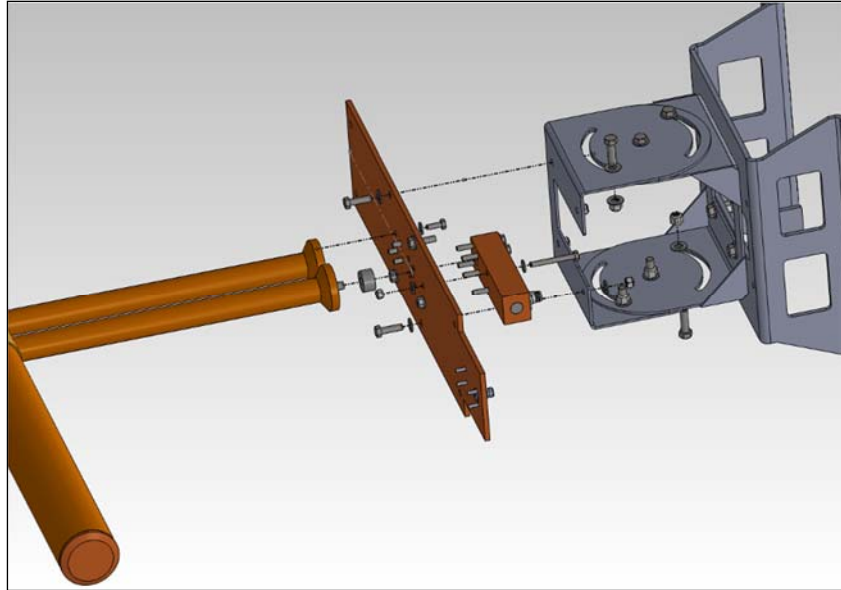


Figure 3-8 Radome assembly

2. Attach the antenna to the tower with 4 U-bolts at a height of 25 feet from the ground.
3. Ensure proper antenna alignment such that it is within $\pm 5^\circ$ of approach path.
4. Attach coaxial cables for transmit and monitor to the antenna.
5. Route the cables down the tower and secure to the tower.



Figure 3-9 Localizer Transmit Antenna



Figure 3-10 Localizer (top) and glide slope antenna attached to the uplink tower

3.5.5 High-gain localizer transmit antenna installation:

- h. Assemble the localizer antenna by matching the element color bands to the mast color bands. Note the color bands on the poles in Figure 5-9, Localizer Antenna Components.



Figure 3-11 Localizer Antenna Components



Figure 3-12 Localizer Assembly Example

- i. Install the monitor and transmitter assembly by opening both clamps and inserting the unit across the red banded element. The unit will seat on the mast alignment screw and be secured using the two thumb screws.



Figure 3-13 Localizer Monitor (MON) and Transmitter (TX) Assembly and Mast Alignment Screw



Figure 3-14 Localizer Monitor and Transmitter Assembly Thumb Screw

- j. Install the localizer antennas on the mast at a height of 22 feet / 7 meters using the localizer mounting assembly installed halfway between the ob-light and the guying ring. The antenna is attached using the push pins on both sides and sliding the antenna over until the pins engage the open holes on the antenna assembly. Ensure the monitor/transmitter assembly is on the lower side of the antenna when mounted to the mast. The primary antenna will be parallel to the front leg of the antenna pointing parallel to the runway

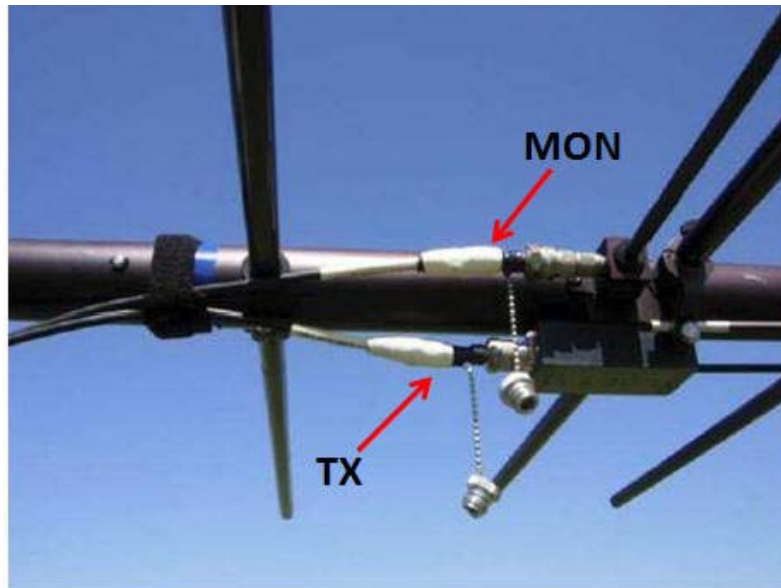


Figure 3-15 Localizer Antenna Connections: MON and TX

- k. Attach the localizer cables to the antenna's LOC TX, LOC MON. The LOC TX cables are Type N connectors and the LOC MON cables are TNC connectors.

3.5.6 High-gain glide slope transmit antenna installation

- l. Assemble the Glide Slope (GS) antenna assemblies. Each assembly consists of two elements and a spreader mount assembly. The connectors of the spreader mount assembly will be oriented downwards. Attach the antenna elements with the monitor ports oriented on the lower side. The elements seat against the set screw on each end of the spreader mount assembly.



Figure 3-16 Unassembled GS Antenna Assy 1 of 2

- m. Attach Glide Slope antennas to the bracket. Each Yagi antenna should slide and lock into place.



Figure 3-17 Attaching Elements to Bracket



Figure 3-18 Alignment Screw on Spreader Bracket



Figure 3-19 Both Elements Attached and Aligned

- n. Attach Glide Slope antennas to mast. The Glide Slope clamp should be attached to the mast approximately 13 ft / 4 m above the ground level.

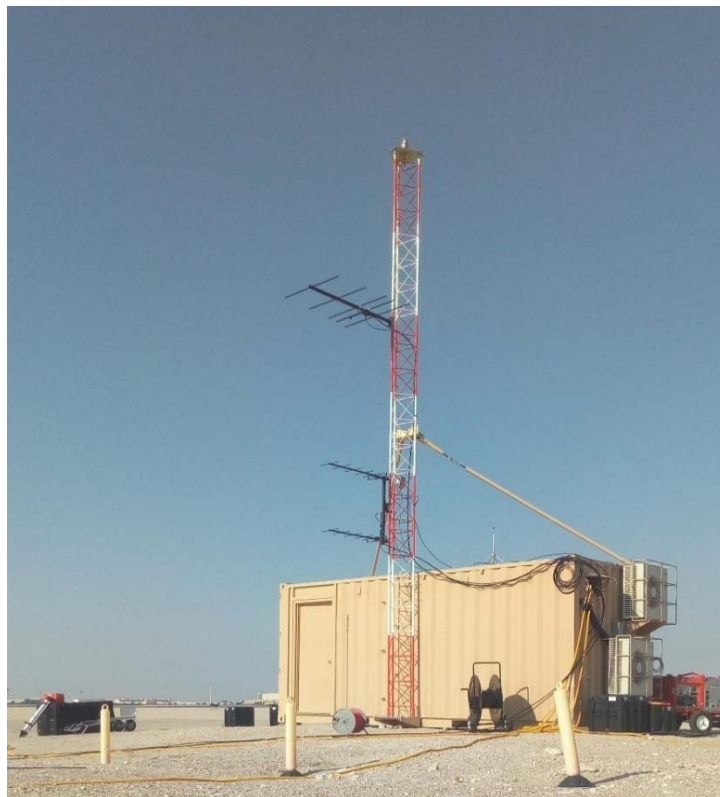


Figure 3-20 - High-gain uplink antennas

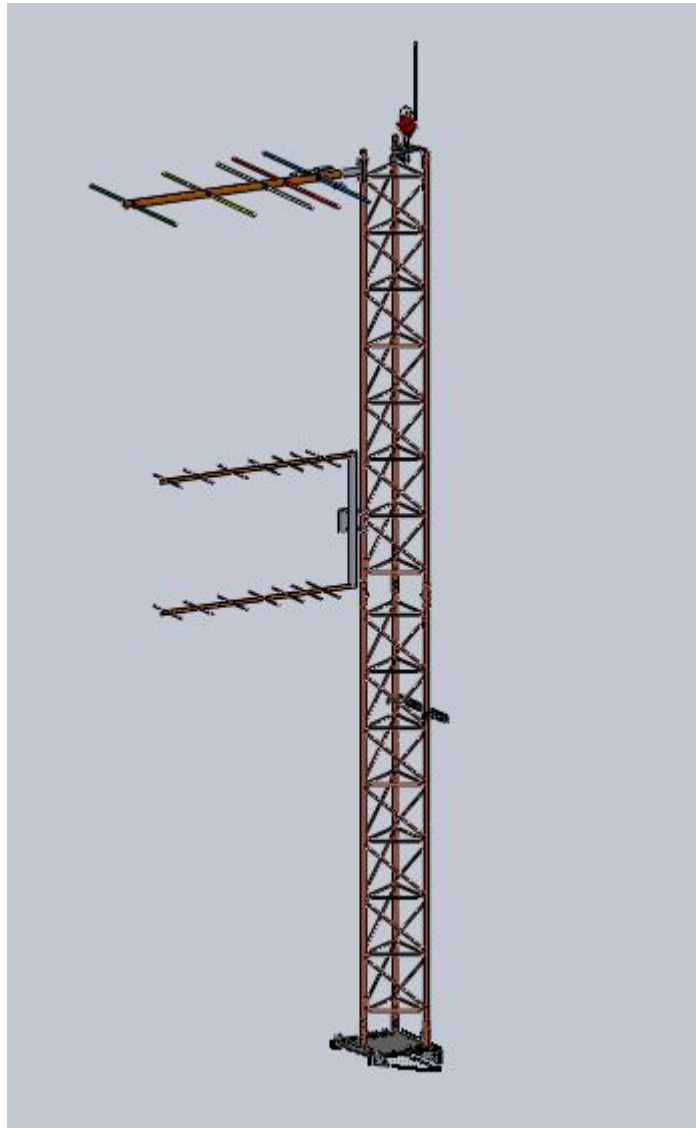


Figure 3-12 - High-gain uplink antennas

3.6 Cal/BIT and ATA Structure Installation

The Cal/BIT structure and the ATA structure [Figures 3-10 and 3-11] are identical in parts and assembly process.

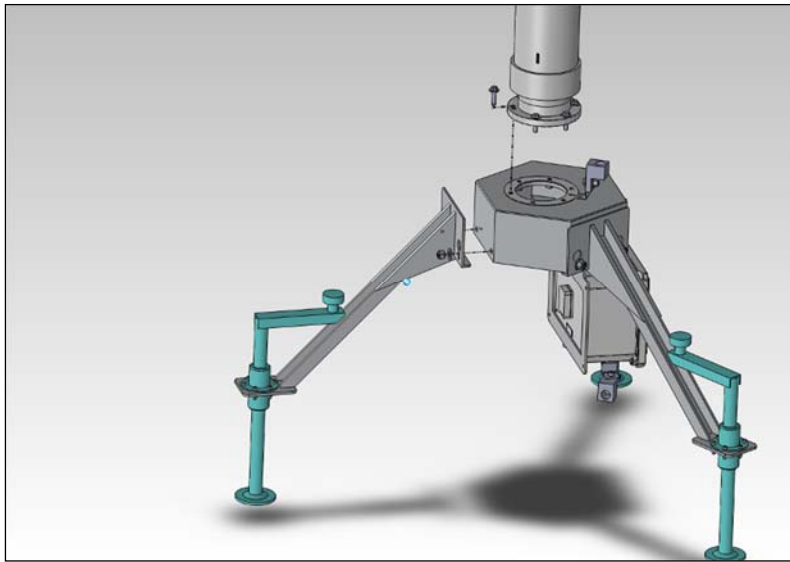


Figure 3-21 Tripod structure assembly (portable version)

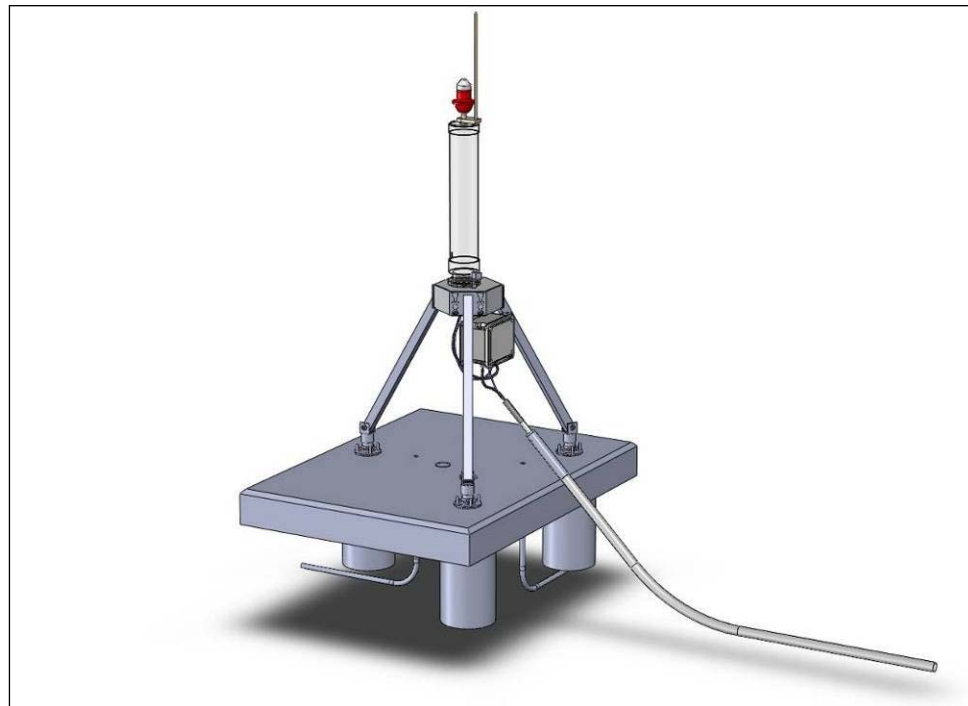


Figure 3-22 Cal/BIT & ATA Structure (fixed install version)

1. Fasten the three steel flanges to the anchor bolts. Include the ground lug on one of the flanges, below the two mounting posts.
2. Apply anti-seize to the threads on the breakable couplings, if it has not already been applied at the factory.
3. Using a pipe wrench, screw the breakable couplings into the posts and the strut bases.
4. Screw the other side of the breakable couplings into the steel flanges. While tightening, be sure to avoid excessive stress to the weak point on the breakable couplings.
5. Level the structure by adjusting the jack screws.



Figure 3-23 Antenna Tripod support leg

3.6.1 Cal/Bit Electronics Installation

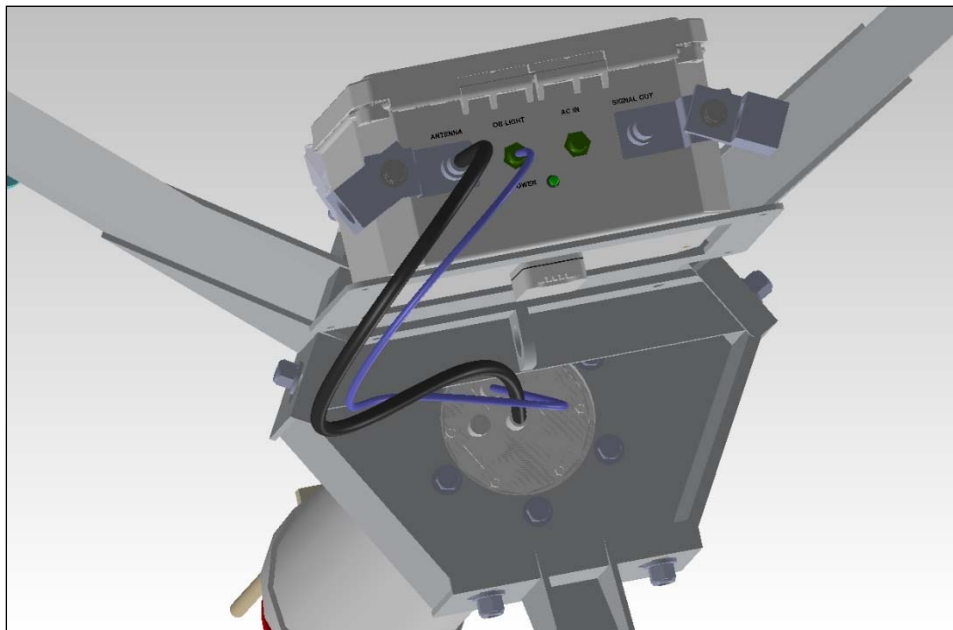


Figure 3-24 Cable connections

1. Bolt the Cal-Bit electronics onto the mounting bracket.
2. Mount the termination assembly to the rails on one of the mounting posts.
3. Mount the Cal/BIT electronics to the mounting posts, ensuring that the 6 AWG grounding cable with an LA lug is attached to the transmitter ground lug.

(Refer to site construction and termination assembly drawings for the following steps)

4. Route the fiber optic/ power cable from the Cal/BIT Transmitter through the hole and rubber grommet in the bottom of the termination assembly.
5. Route fiber and power cables from the Shelter through the conduit access in the bottom of the termination assembly and follow instructions for fiber optic termination.
6. Connect power wires from the base and Cal/BIT fiber optic/power cable to the power termination block. Wrap three inches of the red ground with green electrical tape.
7. Route and connect the other end of the Cal/BIT 6 AWG ground cable from step 3, through the bottom of the termination assembly to the termination ground block.
8. Route and connect another 6 AWG ground cable through the bottom of the termination assembly to the termination ground block. Connect the other end of this ground wire to the 4/0 earth electrode system using an exothermic weld.
9. Connect the fiber lines from the fiber/power cable to one side of the ST bulkheads within the fiber enclosure.
10. Connect the fiber cable with newly installed terminations from step 5 to the other side of the corresponding ST bulkhead connections in the fiber enclosure.

3.6.2 Cal/BIT and ATA Antenna Installation

1. Ensure that the antenna is pointing in the proper direction. For the Cal/BIT, the antenna bore-site should be directed towards the ESA. For the ATA, and if a directional antenna option has been selected, the antenna bore-site should point towards the center of the service volume. Bolt the antenna to the mounting base through the holes in the base of the antenna, installing a grounding plate coated with anti-oxidation compound with tab between antenna base and mounting base.
2. Attach the 32S down-conductor cable with an exothermic-welded LA lug to the tab of the antenna grounding ring.
3. Exothermically weld the bottom end of the 32S down-conductor cable to the designated 4/0 grounding stub that comes from the ground electrode system.
4. For the Cal/BIT, connect the antenna coaxial cable to the antenna and Cal/BIT Transmitter, and seal the connection per the coaxial weatherproofing procedure. Secure the cable to the structure. For the ATA, the antenna coax is a 7/8 inch low-loss cable that connects to the ASA medium RF channel.
5. Install the obstruction light cable to the antenna and seal the connection.
6. Install the air terminal clamp and air terminal with anti-oxidation compound on conductive metal surfaces.

3.7 ASA Installation

When installing the ASA [Figure 3-4], it is important to position the reference antenna of the ASA so that it is farthest from the ESA and position the high-resolution antenna closest to the ESA.

The reference antenna receives the ob-light and is positioned outboard of the adjacent low antenna.

3.7.1 ASA Structure Installation

1. Fasten the four steel flanges to the ASA beam using supplied hardware.

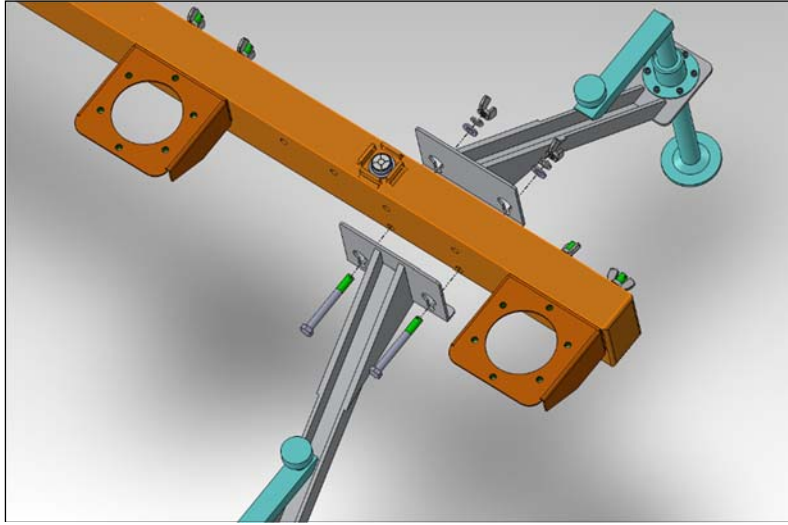


Figure 3-25 ASA leg attachment points

2. Apply anti-seize to the threads on the breakable couplings.
3. Use a vise and pipe wrench to screw the breakable couplings into the two posts and the two strut bases.
4. Screw the other side of the breakable couplings into the steel flanges. While tightening, be sure to avoid excessive stress to the weak point on the breakable couplings.
5. Raise the vertical supports to the necessary height; bolt the left and right interface plates onto the two mounting posts.
6. Bolt angles loosely onto the left and right interface plates.
7. Insert the two struts between the ears of the strut bases and bolt loosely at the proper height.
8. Insert a bolt through each angle, and then screw a nut onto each bolt.
9. Take those same bolts and screw them into the two struts.
10. Tighten the bolts that attach the angles to the interface plates.
11. Level the structure, fore and aft, by tightening or loosening the bolts that go through the angles and into the struts.
12. Once level, tighten the nuts onto the angles.
13. Tighten the bolts that fasten the struts to the strut bases.

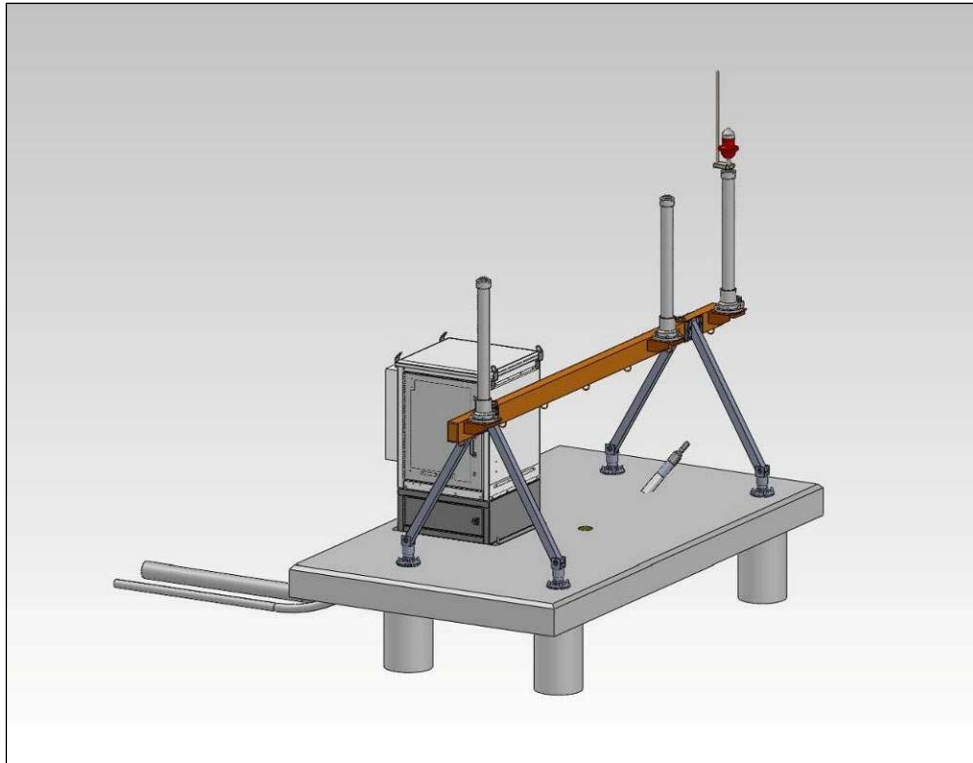


Figure 3-26 ASA Structured

3.7.2 ASA Antenna Installation

3.7.3 Air terminal Installation – may already be installed depending on TLS model number

1. Install the air terminal assembly base onto the pipe and tighten.
2. Remove the plug on top of the antenna farthest from the runway.
3. Using thread sealing compound, install the pipe and pipe cap. Use of thread sealing compound on both ends of the nipple will ensure that water does not get into the antenna. Use counter torque on antenna wrench flats when installing nipple/cap to prevent damage to antenna radome.

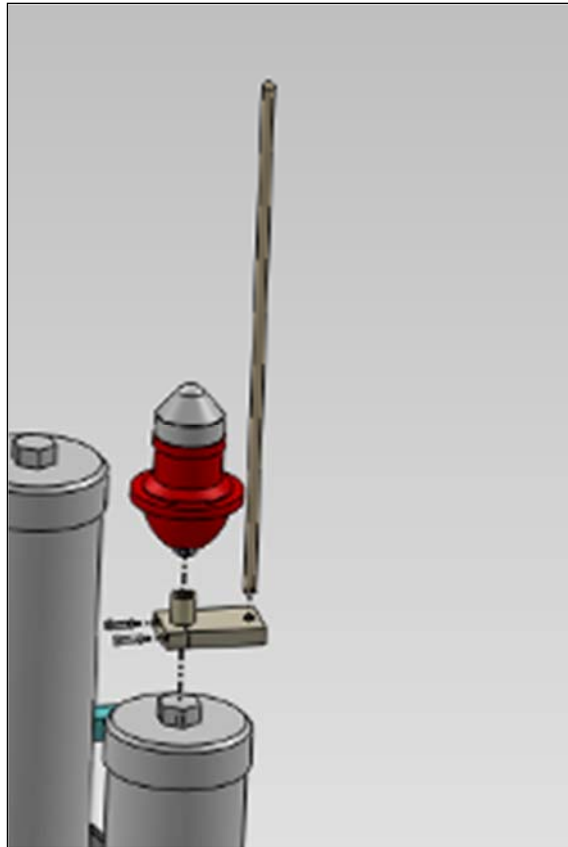


Figure 3-27 Air terminal installation

4. Install the air terminal assembly base onto the pipe with anti-oxidation compound on conductive metal surfaces and tighten.
5. Ensuring that the antennas are pointing in the proper direction (main beam mark on the antenna towards the approach direction), bolt the antennas to the mounting bases through the holes in the base of the antenna. When mounting the antennas with an obstruction light and/or lightning air terminals (antennas at each end of the assembly), install a grounding plate with tab between each antenna base and mounting base.

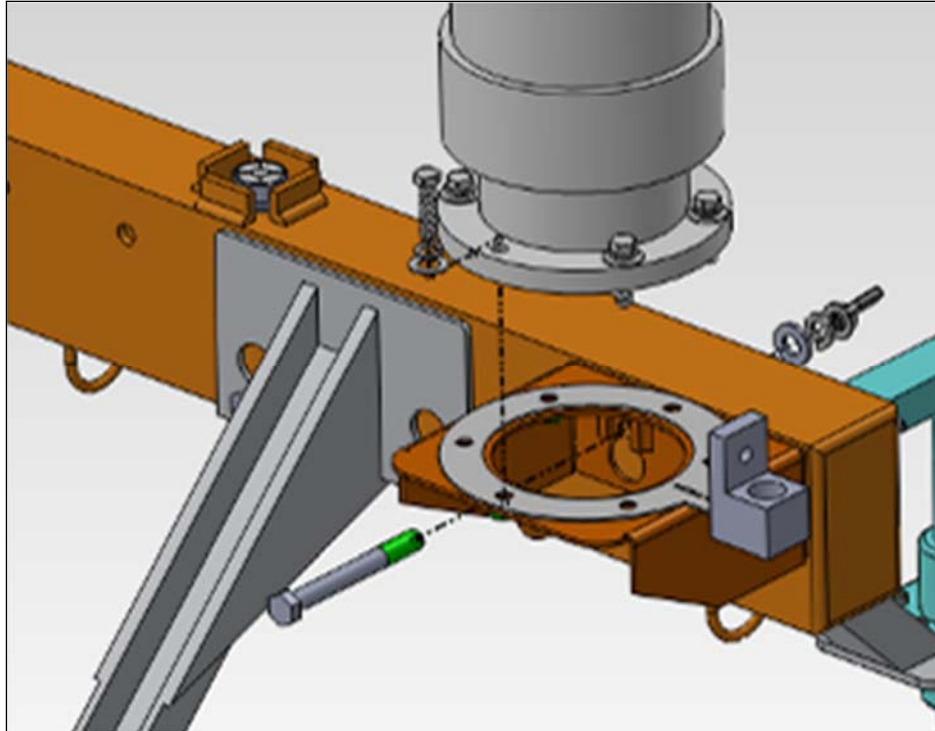


Figure 3-28 Antenna ground ring attachment

6. Attach the 32S down-conductor cable with lug to the tabs of the antenna grounding rings on the Reference and High antennas (each end).
7. Exothermically weld the bottom end of both 32S down-conductor cables to the designated 4/0 grounding stub that comes from the ground electrode system.
8. Install the obstruction light power cable to the antenna with the light installed and seal the connection.
9. Route the obstruction light cable to the termination box and connect to power as follows:
 - **Black** wire to (HOT)
 - **Red** wire (with green tape) to Ground
 - **White** wire to (Neutral)

3.7.4 ASA Sensor Enclosure Installation

1. Route fiber and power cables from the shelter through the conduit access in the bottom of the termination assembly and follow instructions for fiber optic termination.
2. When the foundation requires conduit for the antenna RF cables, ensure that the RF cables are installed and positioned to extend at least 12 to 18 inches above the conduit under the location where the sensor enclosure will be placed.
3. Coil excess RF cable and wire tie to the tower. (It is important that each length of excess cable be exposed to the same thermal changes as the others to minimize phase difference that could result from changes in thermal expansion.)
4. Ensure that each antenna is connected to the correct sensor enclosure connector (Reference, Low, High and ATA to Medium) and seal the connection per the coaxial weatherproofing procedure.

5. Position the sensor housing onto the foundation "J" bolts and secure with nuts and washers. Take care not to damage the fiber optic connectors when positioning the enclosure.

(Refer to termination assembly drawings for the following steps)

6. Connect power wires from the base and ASA fiber optic/power cable to the power termination block. Wrap three inches of the red ground from the sensor fiber/power cable with green electrical tape.
7. Route and connect the other end of the ASA 6 AWG ground cable from step 5, through the bottom of the termination assembly to the termination ground block.
8. Route and connect another 6 AWG ground cable through the bottom of the termination assembly to the termination ground block. Connect the other end of this ground wire to the 4/0 earth electrode system using an exothermic weld.
9. Connect the fiber lines from the fiber/power cable to one side of the ST bulkheads within the fiber enclosure.
10. Connect the fiber cable with newly installed terminations from step 7 to the other side of the corresponding ST bulkhead connections in the fiber enclosure.

3.8 ESA Installation

3.8.1 ESA Tower Assembly

1. Assemble the three tower sections using the supplied hardware.
2. Attach the top and bottom mounting plates to the tower using supplied hardware.
3. Ready the guy wire hardware and hinge plate for the ESA tower. Cut guy wires to 10 to 15 percent oversize lengths, attach the wire thimbles to the turnbuckles, and secure to the guy wire using a wire grip. Refer to site-specific construction drawings for recommended guy wire lengths
4. Attach two out of the three upper guy wires to the j-bolts (concrete option) or augers and lift the tower into position using a crane or winch system.
5. Attach the turnbuckle end of the guy wires to the three anchor eyebolts.
6. Tighten the turn buckles to hold the tower upright.
7. Level the tower by tightening or loosening the turnbuckles using the theodolite as a reference as shown in the figure 3-30 below. Scan vertically on the edge of a tower leg with the theodolite eyepiece and adjust guywire tension until the tower is vertical. To do this, zero the theodolite azimuth at the bottom of the tower and then scan vertically to the top of the tower without changing the azimuth setting. Adjust guy wire tension to move the tower to the reticle of the theodolite eyepiece. Repeat for each axis by placing the theodolite to the runway side of the tower and then in front of the tower (approach side). Repeat on the runway side of tower again if necessary.
8. Check the tension on each ESA tower guy wire with the guy wire tension tool:
 - Top guy wires tension should read 660 lbs. $\pm$ 60 lbs.
 - Bottom guy wires tension should read 400 lbs. $\pm$ 40 lbs.
 - Record validation of proper tensioning of the tower guy wires.
9. Attach the bottom antenna mounting bracket to the tower using 4 U-bolts so that the top of the bracket is approximately 18 inches from the bottom of the tower.

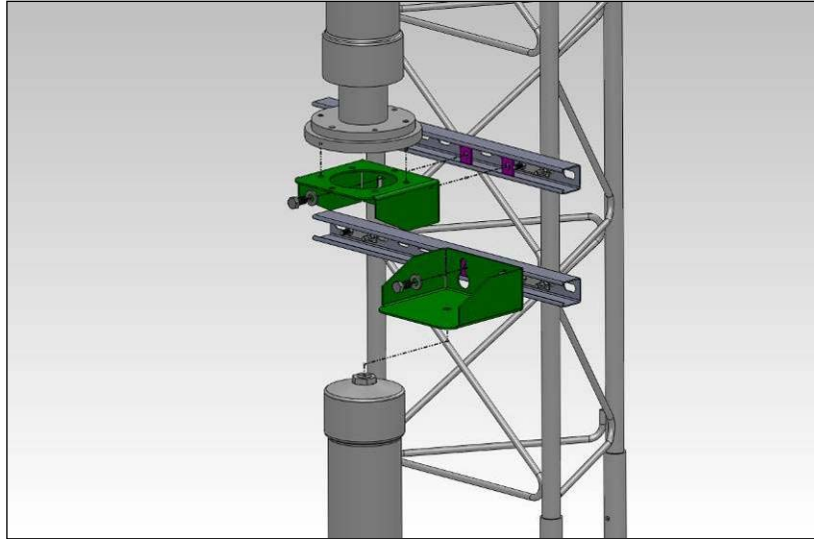


Figure 3-29 ESA Antenna Brackets

10. Insert the antenna mount bolts through the bottom mounting brackets and secure with a rubber O-ring.
11. Bolt the tripod to the bottom mounting bracket.
12. Bolt the antenna to the tripod.
13. The plug in the top of the antenna should then be removed and replaced with the extension to accommodate the upper mounting bracket. Use of thread sealing compound on both ends of the extension will ensure that water does not get into the antenna.
14. Place the bottom nut, the fender washer, and the six-sided washer on the extension.
15. Place the upper mounting bracket in position and secure to the tower using 2 U-bolts.
16. Install the second six-sided washer, fender washer, and nut on the extension, but do not tighten.
- 17.
- 18.
- 19.



Figure 3-30 ESA Antenna Attachment Detail

20. Mount the remaining three antennas by attaching the bottom mounting bracket as close as possible to the top mounting bracket of the antenna beneath.
21. Attach the 32S down-conductor cable with an exothermic welded LA lug to the obstruction light mounting plate. Run and tie the ground cable down the tower leg. A conductive bond is required on every section with anti-oxidation compound, and a secure connection to the tower every 3 feet.
22. Wiring for power to the obstruction light is to be run in conduit and secured to the tower.

CAUTION

Ensure that personnel in the area do not stand under the raised load.

Use of a climbing safety system is required when installing obstruction lights and air terminal on the ESA tower. Always use a two-clip method with climbing harness to ensure constant attachment to the tower.

23. Install High-Visibility guy-wire markers on each of the guy wires.
24. Exothermically weld the bottom end of the 32S down-conductor cable to the designated 4/0 grounding stub that comes from the Ground Electrode System.
25. Power to the obstruction light is to be connected before nightfall on the day the tower is installed.
26. Record validation of proper installation of the ESA Tower.

3.8.1.1 Tilt Sensor Installation

Tilt sensors monitor the stability of the ESA tower through a measurement for tower tilt.

1. Install the tilt sensor mounting plate inside the ESA tower between 60 and 72 inches from the bottom of the tower. Mount to each of the tower legs using U-bolts.
2. Ensure that the plate is level using a torpedo level on each of the three sides of the plate.
3. Mount the tilt sensors to the mounting plate with the provided hardware. Standing behind the tower facing the runway threshold, the parallel tilt sensor cable connector will face to the right, and the perpendicular tilt sensor cable connector will face toward the service volume and the general direction of the runway threshold. Refer to Figure 3-10 for proper tilt sensor orientation.
4. Record validation of proper installation of the Tilt Sensor Assembly.

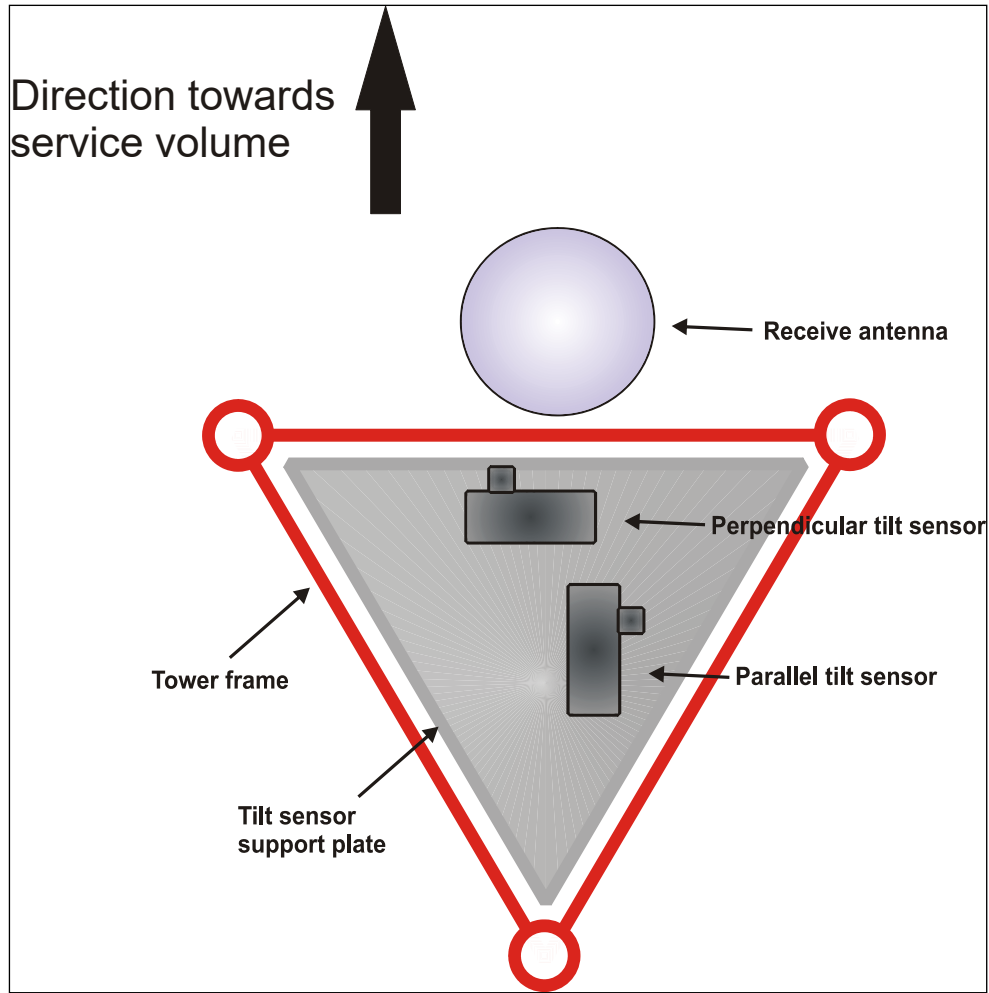


Figure 3-31 ESA Tilt Sensor Orientation

3.8.2 AOA sensor installation

Identify the AOA sensor and read the sensor model number label to determine the AOA sensor part numbers included with the system. The part number will be either one of the following:

- 920-00256 or,
- 920-00400

Follow the instructions below for the part number supplied.

3.8.2.1 920-00256 ESA and ASA Sensor Enclosure Installation

1. Route fiber and power cables from the Shelter through the conduit access in the bottom of the termination assembly and follow instructions for fiber optic termination.
2. Attach the coaxial cable to the four antennas and seal each connection per the coaxial weatherproofing procedure.
3. Route the 4 antenna coaxial cables down the tower, securing them to the tower at regular intervals.
4. If the foundation includes conduit for the antenna RF cables, then ensure that the RF cables are installed and positioned to extend at least 12 to 18 inches above the conduit under the location where the sensor enclosure will be placed.

5. Coil excess RF cable and wire tie to the tower. (It is important that each length of excess cable be exposed to the same thermal changes as the others to minimize phase difference that could result from changes in thermal expansion.)
6. Ensure that each antenna is connected to the correct sensor enclosure connector (Reference, Low, Medium and High) and seal the connection per the coaxial weatherproofing procedure.
7. Position the sensor housing onto the foundation "J" bolts and secure with nuts and washers. Take care not to damage the fiber optic connectors when positioning the enclosure.

(Refer to termination assembly drawings for the following steps)

8. Connect power wires from the base and ESA fiber optic/power cable to the power termination block. Wrap three inches of the red ground from the sensor fiber/power cable with green electrical tape.
9. Route and connect the other end of the ESA 6 AWG ground cable from step 4, through the bottom of the termination assembly to the termination ground block.
10. Route and connect another 6 AWG ground cable through the bottom of the termination assembly to the termination ground block. Connect the other end of this ground wire to the 4/0 earth electrode system using an exothermic weld.
11. Connect the fiber lines from the fiber/power cable to one side of the ST bulkheads within the fiber enclosure.
12. Connect the fiber cable with newly installed terminations from step 6 to the other side of the corresponding ST bulkhead connections in the fiber enclosure.

3.8.2.2 920-00256 ESA Sensor Electronics Installation

1. Mount the sensor electronics to the enclosure chassis mounting brackets, ensuring that the 6 AWG grounding cable with an LA lug is attached to the sensor ground lug.
2. Connect the RF cables via SMA connectors to the enclosure bulk head N-Type connections. Take care to match each RF channel by antenna function e.g. *Reference, Low, Medium and High*. Figure 3-5 and Figure 3-6.
3. Connect short "pigtail" fiber optic network transmit and receive fibers to the sensor and the ST bulkhead. Take care to not swap the network fibers (If fibers become swapped, TLS Shelter software will not be able to ping the sensors and will be stuck in *Initializing* mode). At the Shelter, these fibers attach through the fiber optic tray to the Ethernet hub. Figure 3-7
4. Connect short "pigtail" fiber optic T0 cable from the ST bulk head to the T0 connection. At the Shelter, this fiber connects to the fiber optic splitter attached to T0 on CPU1. Figure 3-7.

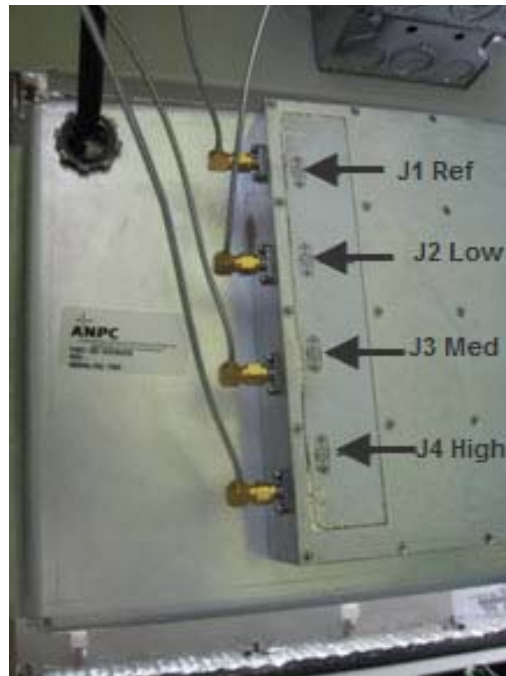
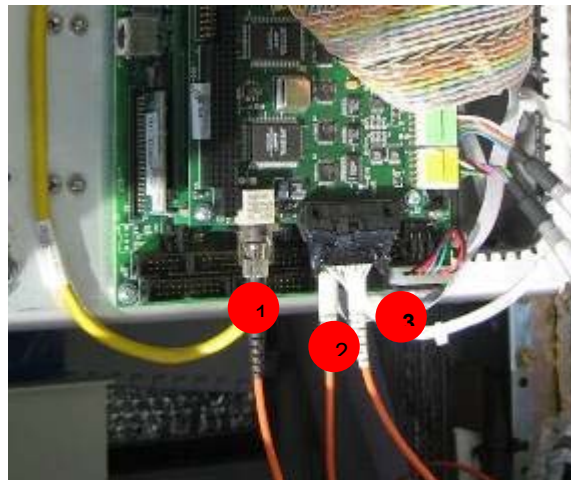


Figure 3-32 PRM Cable Connections



Figure 3-33 Cable and Ground Lug Attachment Points on Sensor



1. T0
2. Transmit
3. Receive

Figure 3-34 Fiber Optic Attachment Points on Sensor Electronics

3.8.2.3 920-00400 ASA and ESA Sensor Electronics Installation

1. For the ESA: Use the provided c-channel and u-bolts to attach the sensor supports to the tower approximately 4 feet above the base plate level of the tower. The sensor will be on the side of the towers away from the side with the antennas attached. Fasten the sensor to the c-channel as shown using four bolts.

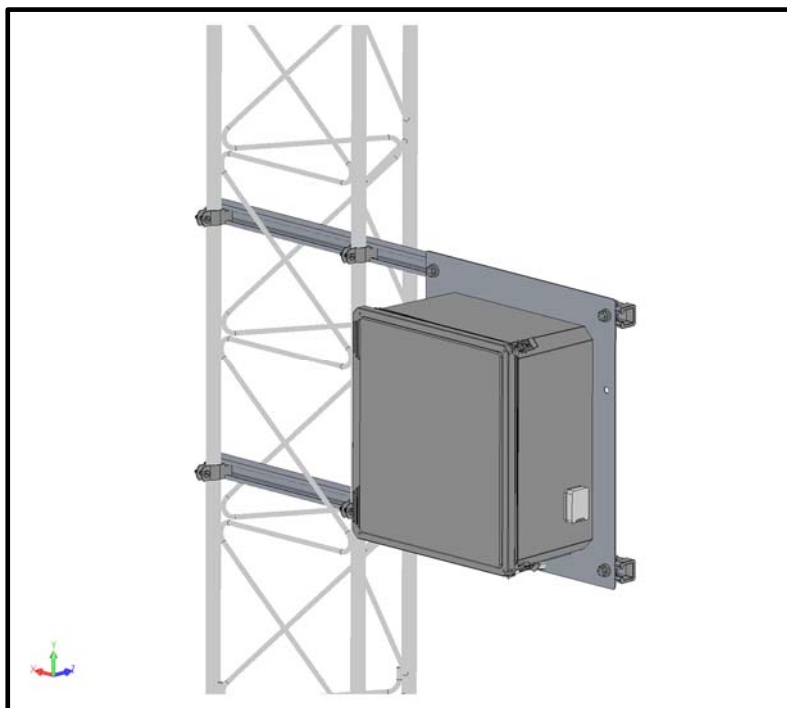


Figure 3-35 – ESA electronics installed

2. Route fiber and power cables from the Shelter through the conduit access in the bottom of the junction box assembly and follow instructions for fiber optic termination.
3. Attach the coaxial cable to the four antennas and seal each connection per the coaxial weatherproofing procedure.
4. Route the 4 antenna coaxial cables down the tower, securing them to the tower at regular intervals.
5. Coil excess RF cable and wire tie to the tower.
6. Ensure that each antenna is connected to the correct sensor connector (Reference, Low, Medium and High) and seal the connection per the coaxial weatherproofing procedure.



Figure 3-36 – AOA sensor connections

(Refer to termination assembly drawings for the following steps)

7. Connect power wires from the base and ESA fiber optic/power cable to the power termination block in the junction box.
8. Route and connect the other end of the ESA 6 AWG ground cable, through the bottom of the termination junction box assembly to the termination ground block.
9. Route and connect another 6 AWG ground cable through the bottom of the termination assembly to the termination ground block. Connect the other end of this ground wire to the 4/0 earth electrode system using an exothermic weld.
10. Connect the fiber lines from the fiber/power cable to one side of the ST bulkheads within the fiber enclosure of the junction box.
11. Connect the fiber cable with newly installed terminations from step 6 to the other side of the corresponding ST bulkhead connections in the fiber enclosure.
12. For the ASA: locate the 8 inch u-bolts intended to fasten the AOA sensor electronics to the ASA beam. The ASA sensor should be fastened to the beam using the supplied u-bolts at the midpoint between the low resolution antenna and the high resolution antenna.

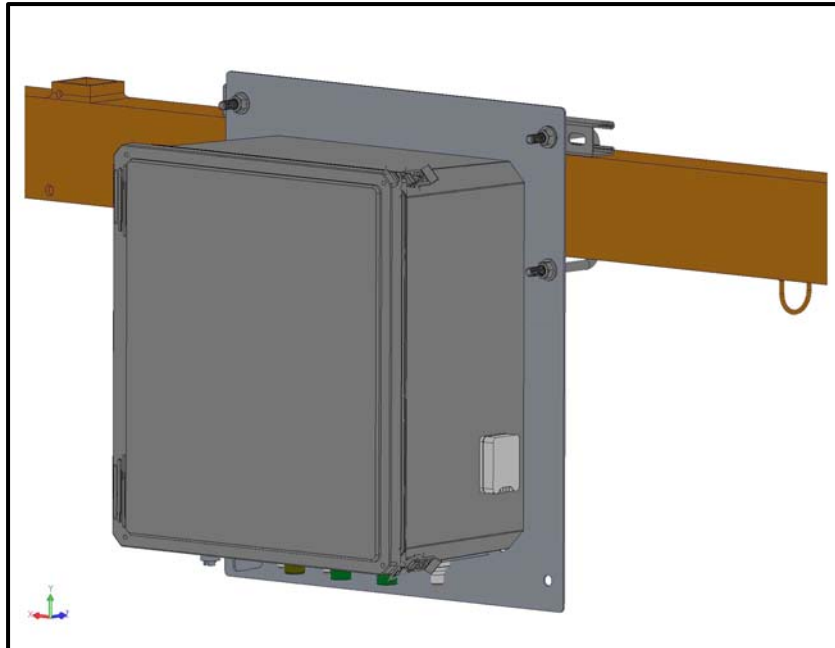


Figure 3-37 – ASA electronics installed

3.9 Interrogation Antenna Structure Assembly

3.9.1 Structure Installation

1. Assemble the tripod base as shown
2. Bolt the pipe flange to the assembly foundation.
3. Tighten the antenna mount onto the tripod so that the axis of the mount is in line with the runway centerline to within ± 5 degrees. Final orientation will be determined after installing the antennas.

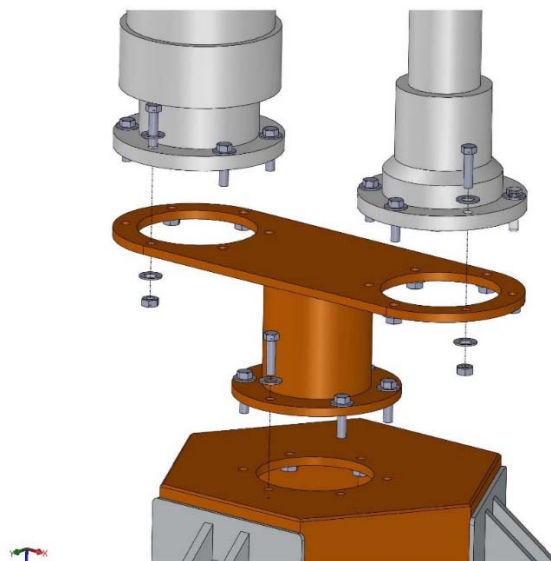


Figure 3-38 Omni-directional Interrogation and Suppression Antenna Assembly

1. Attach the P2 antenna at the opposite side of the mount, so that the MAIN BEAM mark is 180 degrees from the approach centerline.

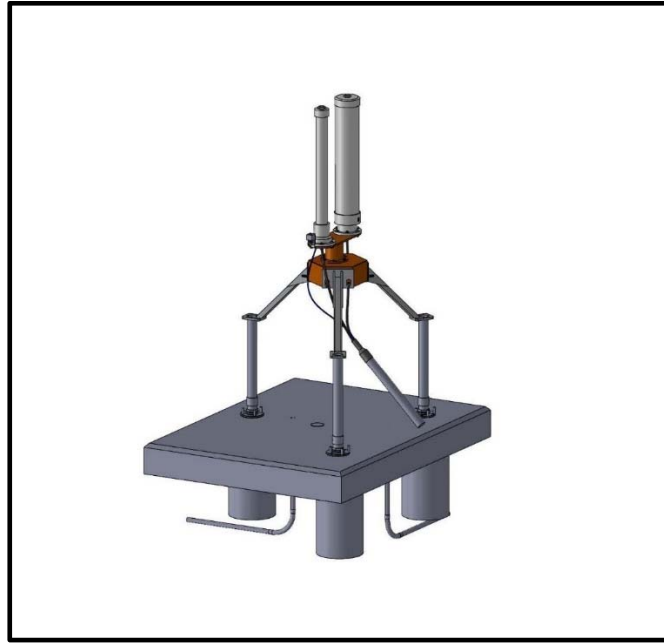


Figure 3-39 Interrogator antenna tripod structure

3.9.2 Interrogation Antenna Installation

2. The Main beam of the directional interrogator antenna must point toward the approach. Note that this instruction does not apply to the omni directional interrogation antenna. If equipped with the suppression option, the main beam of the suppression antenna must point away from the approach. .
3. Main beam aiming survey: Survey a stake parallel to the runway centerline at the same distance from runway centerline as the interrogator. The stake must be placed 100 feet away from the interrogator to use as a siting reference to aim the main beam of the interrogator antenna and the side lobe suppression main beam as applicable.



Figure 3-40 "Main Beam" marking on Antenna

4. Connect coaxial cables to the two antennas, and seal the connection per the coaxial weatherproofing procedure.

5. Position the antenna assembly with the interrogation antenna main beam within 5 degrees of the same heading as the extended runway centerline, use the stake from the “Main beam aiming survey” procedure step above as the aiming point for the main beam.

3.10 Fiber Optics

3.10.1 Testing the Spliced fiber optic cable for TLS (not needed for Transportable TLS)

1. After all splicing has been completed, one technician should attach the light optical source to the first pigtail at one end of the fiber optic cable using a calibrated test cable and ST bulkhead. For the sensor multimode fiber, set the Source for an 850 nm wavelength output.
2. A second technician should attach the light test meter to the other end of the first fiber optic cable being tested. Set the light meter to read the 850 nm light. Communicate and coordinate with radios to do this test.
3. Verify an acceptable loss using the light test meter. Verify a loss of less than 5 dB for the total length of any fiber including all ST connectors.
4. Move the source and meter to the second fiber and verify loss and repeat until all fibers are verified.
5. If any splice fails to meet the acceptable loss limits, it must be re-spliced and re-tested.
6. Record the measured loss for each fiber.
7. If the RCU is installed with the fiber optic option, continue with the steps below to verify the fiber optic connections for the RCU.
8. For the single-mode fiber used to connect the RCU, use the light source and meter measurement settings for a 1300 nm light signal.
9. Verify an acceptable loss using the light test meter. Verify a loss of less than 5 dB for the total length of any run.
10. If any splice fails to meet the acceptable loss limits, it must be re-spliced and re-tested.
11. Repeat steps 1 through 8 for all fibers spliced.
12. Record the measured 1300 nm loss for the fiber optic.

3.11 Remote Control Unit Installation

RCU(s) will be installed at a site-specific location that provides access to the operators, and can be serviced by one of the optional TLS connection methods. Because this connection is site specific, the ANPC site drawing will be used to provide details for installation of this connection. Up to four RCU may be installed for use with one TLS. Each RCU connectivity to TLS can be one of the following options.

3.11.1 Drawings: General Reference Only

The following RCU installation option drawings and descriptions are for general reference only. Actual drawing for site-specific RCU installation option will be made available for each site.

WARNING

Ensure that all RCU installation options have the required surge suppression UPS that modems are secured or strapped down.

Record the type of RCU connection implemented and validation of proper installation. The information that follows presents the current acceptable options for the RCU link implementation.

3.11.2 RCU Direct Ethernet Option

This option is for connecting a RCU with the TLS Shelter equipment when the RCU is positioned within 300 feet. Interconnection is made via a cable terminated with RJ-45 style connectors. One end of the cable plugs into any of the unused RJ-45 ports on the rear of the Ethernet hub. The other end of the cable plugs into the Ethernet port of the RCU computer. This RCU connection is presented in Figure 3-12.

3.11.3 RCU Wireless Ethernet Option

The RCU Wireless Ethernet option [Figure 3-12] uses a point to point wireless connection to link the RCU from a remote site to the Shelter. The wireless option is recommended only if the RF radio is configured as an Ethernet bridge. Do not use dial up connections because of the high maintenance associated with reestablishing such connections after an interruption.

Wireless modems bridge the Ethernet switch at the base shelter and the remote computer Ethernet switch or port. . A power block is supplied with the modems. Each of the modems is connected directly to the antenna to achieve the most RF gain and maximize connectivity between the RCU and the shelter.

The RCU modem antenna can be located inside (near a window) or outside on the uplink structure. The RCU antenna directly connects to the modem with a coaxial cable.

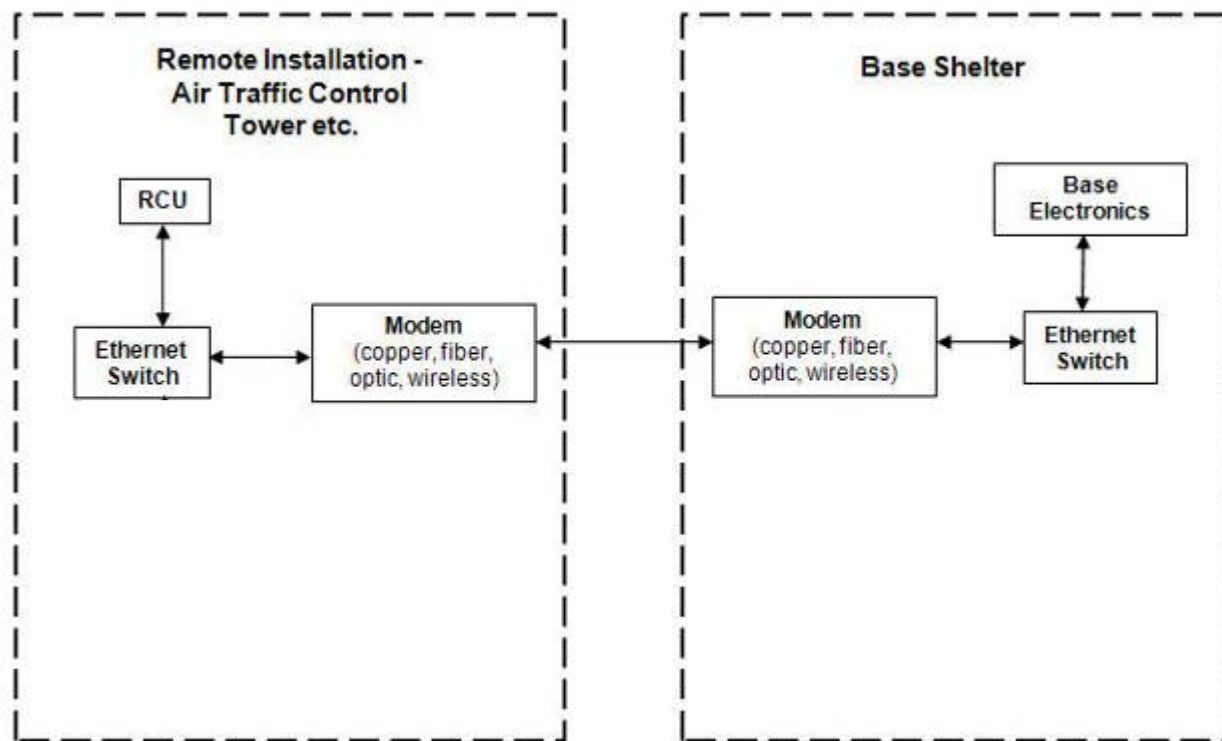


Figure 3-41 RCU Configuration

3.11.4 RCU Ethernet with Media Converter (fiber optic)

This option is used to connect the RCU from a remote site to the shelter via fiber optic or twisted pair media. Ethernet media converters are installed at the shelter electronics rack and at the RCU.

3.11.4.1 Fiber

Fiber: Two single mode fibers terminated with SC connectors attach to the RCU's media converter. A Ethernet cable terminated with RJ-45 connectors, jumpers between the media converter and the RCU

Ethernet port. At the Ethernet hub, in and out fibers connect to the media converter. An Ethernet cable connects from the media converter card to one of the unused hi-speed RJ-45 ports 1-12 on the Ethernet switch hub. When the connection is with fiber-optic lines, there are no data-line surge suppression elements.

3.11.4.2 Twisted Pair

Twisted Pair: Twisted pair lines are connected to the RCU's media converter. A short Ethernet cable, terminated with RJ-45 plugs, connects between the media converter and the RCU Ethernet port. At the Ethernet hub, in and out fibers connect to the media converter. An Ethernet cable connects from the media converter card to the any one of the unused hi-speed RJ-45 ports 1-12 on the Ethernet hub.

3.12 Critical Area Signage Installation

Coordinate with the airport management and ATC for the install of critical area warning signs at all locations where roads and taxiways enter the TLS glide slope, localizer and Cal/BIT critical areas. Also coordinate with airport facilities for marking "Hold Lines" on taxiways.

3.13 Ground to Air Radio

The Ground to Air radio equipment is provided as a kit defined in the Comm Radio Assembly drawing (included in Base Shelter Assembly). This kit contains the radio, antenna, cables and RF lightning protection device.

1. Install the communications antenna to the uplink tower on corner of the shelter using the installation hardware.
2. Attach one comm antenna coax between the interface panel on the shelter and the antenna. Attach the other comm antenna coax between the interface panel and the radio.
3. Use the site Unicom frequency to perform a radio check to the site Unicom operator.



Figure 3-42 Air-band communications radio

SECTION 4: CONFIGURATION

4.1 Quality Inspection

Prior to initial system startup, all components should be inspected to ensure proper installation (see Appendix C, doc. #020-00074). Inspect each assembly and record validation of proper installation, and the quality of the workmanship used in the completion of the physical installation on the TRR.

Special care should be taken to look for and to repair inadvertent damage to painted surfaces that may have occurred during construction. Record the validation of proper installation for each component.

4.2 Safety Inspection

Prior to initial system startup, all major components should be inspected to ensure that there are no safety hazards related to the system installation. Walk to each assembly and take special care to look for sharp edges, construction materials on the site, guy wire hazards, tripping hazards etc. Record the completion of the safety inspection of the site.

4.3 Configuration Audit

Before shipping the system and prior to initial system startup, inspect all major components to ensure that the installed configuration accords with system configuration documentation. Review assembly level As Designed As Built (ADAB) paperwork and verify a match to physical installation. Record validation of proper configuration.

4.4 Final Precision Survey

After the TLS sites have been installed, the ASA, ESA, ATA, Cal/BIT and Interrogation antennas must be surveyed. The survey is conducted relative to the runway threshold, the actual latitude and longitude for each point is not needed to operate TLS, the latitude and longitude is only needed for requests to the regulator to place the equipment on airport property. The survey accuracy should be within 3 feet (1 meter) to honor the Lat/long indicated on the request form. The TLS calibration process (doc 020-00071) eliminates the error of the survey. Note that the elevation is measured and recorded for only the ESA.

WARNING

Coordinate with the airport authority to close the runway for the time required since work will be carried out on and around the runway.

Personnel and Equipment Resources:

- Two installation technicians
- One theodolite with Electronic Distance Meter (EDM), survey staff and prism..
- The MI survey data entry screen requires the units to be in feet. The theodolite should be configured to display distance units in feet. It is not permitted to measure the distance in meters and then use a calculator in Windows or calculator phone app to convert to feet. To reduce the possibility of human error, the theodolite must be used in units of feet.
- Additional 25' tape measure.
- Record verifications on a notepad with a pen or in Table C-3, Precision Equipment Survey

4.5 Equipment Survey

1. Set up Theodolite.

After permission to approach the runway has been granted, Installer #1 should place the theodolite at the centerline, threshold point of the runway. Level the theodolite, and if the theodolite has an EDM capability, set it up for distance measuring. Installer #2 must setup the survey staff height such that the prism is within an inch of the theodolite eyepiece. Do this with the bottom of the staff placed on the pavement next to the theodolite and the prism within a few inches of the eyepiece pivot point.

*Lock the prism staff at this height and DO NOT change throughout the complete final precision survey procedure.

2. Orient Theodolite.

Installer #1 should align the theodolite eyepiece on the centerline of the runway and set the centerline as the zero azimuth measurement on the theodolite. Next, to align the theodolite with the TLS coordinate system, rotate the theodolite in azimuth to a measurement of 180° and then re-zero the azimuth. Measurements from the theodolite will now match the orientation and sign convention shown in Figure 4-1. Measurements may not be input to a phone for record keeping to transfer to the MI after the survey is complete.

3. Survey ASA

Installer #2 must position the staff next to the ASA reference antenna such that the prism is between the theodolite and the antenna. Installer #1 should use the theodolite to measure azimuth and distance to the prism then record the measurements on a notepad with a pen.

Survey ATA

Installer #2 must position the staff next to the ATA antenna such that the prism is between the theodolite and the antenna. Installer #1 should use the theodolite to measure azimuth and distance to the prism and then record the measurements on a notepad with a pen.

4. Survey Cal/BIT.

Installer #2 must position the staff next to the Cal/BIT antenna such that the prism is between the theodolite and the antenna. Installer #1 should use the theodolite to measure azimuth and distance to the prism, and then record the measurements on a notepad with a pen.

5. Installer #2 must position the staff next to the Interrogation antenna - between the theodolite and the interrogation (P1/P3) antenna, and such that the prism is between the theodolite and the antenna. Installer #1 should use the theodolite to measure azimuth and distance to the prism, and then record the measurements on a notepad with a pen.

6. Survey ESA.

Installer #2 must position the staff next to the ESA reference antenna such that the prism is between the theodolite and the antenna and resting on the base plate of the ESA tower. Installer #1 should use the theodolite to measure azimuth, and distance to the prism, and then record the measurements on a notepad with a pen. Flip the eyepiece 180 degrees and turn the theodolite in azimuth 180 degrees so that the elevation measurement can be made using 90 deg for the horizon. Check again the level of the theodolite and adjust if necessary, disregard the azimuth measurement and measure the height of the prism positioned at the ESA reference antenna. Record the raw measurement on a notepad with a pen. Subtract 90 from the measurement recorded. The result is the elevation that must be entered in the MI survey data entry screen for the ESA elevation.

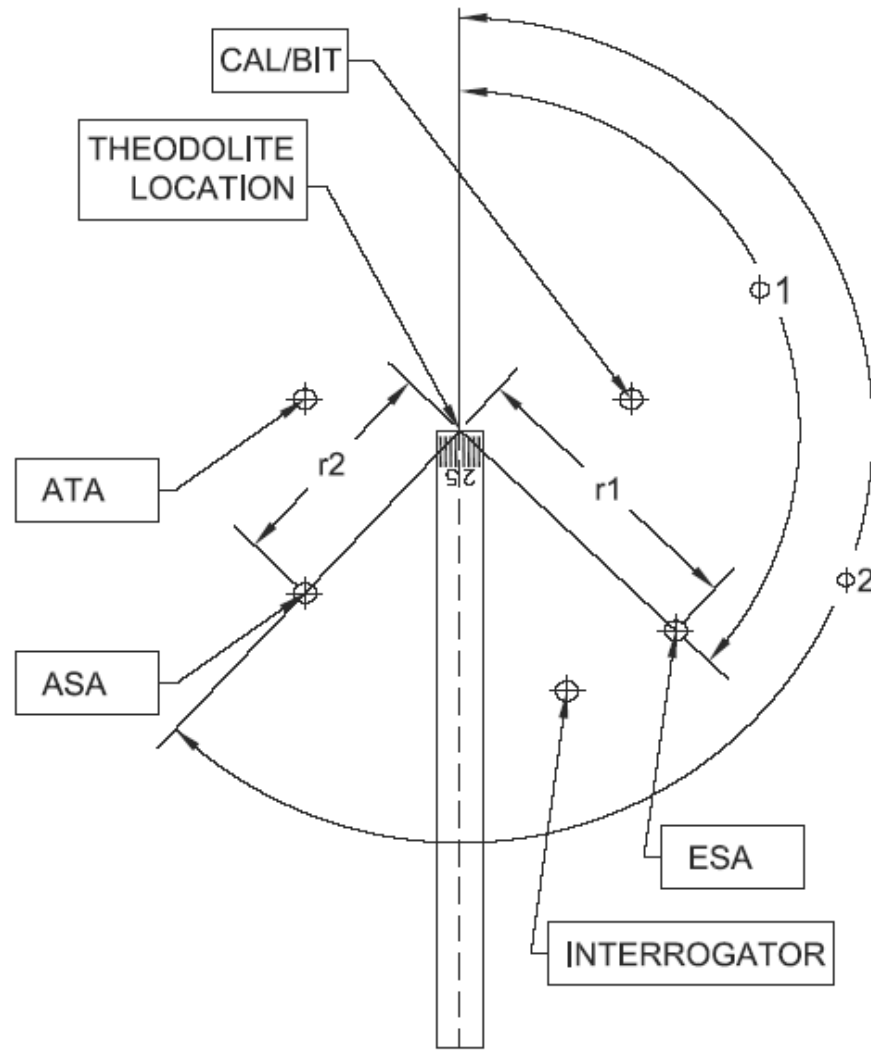


Figure 4-1 Equipment Survey Sketch

4.5.1 FTM Theodolite Position Survey

This procedure will establish the location for the Flight Test Monitor (FTM) theodolite that achieves a theodolite eyepiece height within a foot of the ideal glide path. Provisions must be made in advance to have access to the runway threshold for 30 minutes.

The ground elevation at threshold is likely to differ from the ground elevation at the FTM theodolite location. An approximate setback from threshold must be used to find the ground elevation difference and then the actual setback may be computed.

1. Determine the offset from centerline for the theodolite equipment. Select the offset location to permit theodolite operation during flight-testing without closing the runway. A location on the grass or dirt area within 10 feet of the runway shoulder is usually acceptable. Consult with the airport manager or their designee to ensure that the selected offset may be used. Measure and record this offset, os , from centerline, as it will be needed to correspond to the test approach offset used.

2. Calculate the distance from the survey mark at runway threshold to the FTM theodolite position. This distance is determined by:

$$T_1 = \sqrt{\left(\frac{tch - theo - dz}{\tan(pa)}\right)^2 + os^2}$$

Where:

<i>tch</i>	=	threshold crossing height obtained from FAA Form 8260
<i>theo</i>	=	the theodolite eyepiece height, typically 5.5 feet
<i>dz</i>	=	elevation difference from the threshold to the theodolite
<i>pa</i>	=	glide path angle
<i>os</i>	=	offset from centerline

*Calculate and record the distance T_1 with an initial value of dz set equal to zero.

Example: For a 50 foot crossing height, 5.5 foot eyepiece allowance, a 3.0 degree glide path and an offset from runway centerline of 75 feet, $T_1=852.4$ feet.

3. Request permission from the airport manager or tower to approach the runway for 30 minutes of survey activity. After receiving permission, set up the survey theodolite over the survey mark at runway threshold and set the prism staff height equal to the height of the theodolite eyepiece. Make sure the staff height adjustment does not change after this point. Pace off an approximate distance equal to T_1 from threshold for the survey staff adjacent to the runway shoulder.
4. With the survey theodolite still over the survey mark at runway threshold, sight toward the survey staff location. Using the range measuring capability of the theodolite, adjust the placement of the staff until it is at the distance calculated during step 2.
5. Measure the elevation angle *el*. Convert the measured angle from degrees, minutes and seconds to degrees, keeping in mind that there are 60 minutes per degree and 60 seconds per minute. The sign of dz will be greater than zero if the ground height at the FTM theodolite location is above the runway threshold. Compute dz equal to:

$$dz = \tan(el) \times T_1$$

6. Compute T_2 using the value for dz found during step 5:

$$T_2 = \sqrt{\left(\frac{tch - theo - dz}{\tan(pa)}\right)^2 + os^2}$$

Example: For a 50 foot crossing height, 5.5 foot eyepiece allowance, a 3.0 degree glide path and an offset from runway centerline of 75 feet, with a dz of -3.5 feet, $T_2=918.9$ feet.

7. Measure the distance calculated during step 6 and then mark this location.
8. Repeat step 5. If dz changed by more than 1 foot from the previous measured location, repeat steps 6 and 7 to calculate T_3 using the same formula as for T_2 . T_2 or (T_3 if applicable) will be entered into the FTM software, and the position on the airfield will be marked permanently with a concrete slab and set-nail. Record final distance from threshold for the FTM location.

9. Rotate the theodolite 90° from centerline, toward the side of the runway adjacent to the FTM theodolite position. Measure and record the offset distance determined in step 1 above and mark this point as the theodolite back-sight. Permanent identification markings must be provided at a later time for this back-sight position and the FTM theodolite position determined during step 8.
10. Carry the survey theodolite off the runway and inform the airport authority by phone or radio that operations on the runway are complete.
11. Record validation of proper measurement and permanent marking of the FTM theodolite position.

4.5.2 Measure ESA Antennas

1. Base to Reference.

Measure the distance from the tower base plate where the bottom of the survey staff had been placed in section 4.5 step 7, to the bottom of the antenna flange. Note that the TLS survey data entry screen will later require the distance to the phase center which is 44 inches up from the base of the antenna flange. Add 44 inches to the measured distance from the base to the antenna flange (in Figure 4-2) and record this measured phase center height of the Reference antenna (inches).



Figure 4-2 Reference Antenna Flange to Measure

2. Reference to Low, Medium and High antennas. Measure to the same point used in step 1 for each antenna
 - Measure the distance from the flange of the reference antenna to the flange of the low-resolution antenna. Record this measured Ref - Low distance (inches).
 - Measure the distance from flange of the reference antenna to the flange of the medium-resolution antenna. Record this measured Ref - Med distance (inches).
 - Measure the distance from the flange of the reference antenna to the flange of the high-resolution antenna. Record this measured Ref - High distance (inches).

4.5.3 Mask Angle Measurement.

A theodolite measurement of the ground obstructions or Mask Angle is required because the TLS is a line-of-sight system (equipment needs to have an unobstructed "view" of the aircraft being tracked). A theodolite is used to measure +/- 45 degrees about the approach heading and measure the elevation of obstacles.

1. Set the theodolite up in front of the Interrogation antennas (within 5 feet).
2. Align the theodolite eyepiece on the stake used for alignment of the Interrogation antennas in section 3.9 . Set this heading as the zero azimuth point on the theodolite.
3. Set the elevation of the theodolite and then record the lowest point at which sky is visible. Record this mask angle elevation for 0 degrees azimuth.
4. Rotate the theodolite slowly, measure and record the elevation mask angle every 5 degrees through ± 45 degrees from the runway heading. If there is an obstruction that peaks in between two 5 degree readings, measure and record the peak elevation and record to the nearest 5 degree azimuth point of the obstruction peak.
5. Record the defined mask angle and lowest unobscured azimuth space. This data will determine the Lowest Coverage Altitude during flight inspection.

4.6 Special Tools and Equipment

The special hardware tools and equipment required for configuration of the TLS are listed and described in Table 4-1. Verify that the equipment that is required to have current calibration status stickers are so marked. Record this validation.

Table 4-1 Special Tools and Test Equipment

Description	Function	Calibration Required
Theodolite	System siting and calibration	No
Flight Test Monitor (FTM) equipment (modems, antennas, cables, etc)	System calibration/Flight inspection	No
Portable ILS Receiver (PIR)	Guidance transmission calibration	Yes
Oscilloscope	System Startup and Configuration	Yes
Digital Multi-Meter (DMM)	System Startup and Configuration	No
Crystal Detector	Interrogation transmitter delay meas.	No
Coaxial Attenuators, various	Interrogation and GTU tests.	No

4.7 Switching Keyboard and Monitor Control between CPUs

The keyboard, mouse and monitor in the equipment rack are connected to the CPUs through a KVM switch. To switch to a different CPU, follow the instructions label on the KVM to determine the hotkey and select the desired computer.

The remaining procedures will assume that the user is familiar with switching the keyboard and monitor interface to the different base CPUs.

4.8 Rack Electronics Power On

1. Verify that all power switches on TLS components in the Base rack are in the OFF position prior to applying power.
2. Plug the AC output of the UPS into the circuit breaker panel input at the shelter wall. Disconnect the Shelter UPS power that returns to the rack.

3. Open the UPS distribution breakers for the ESA, the ASA, and the Cal/BIT.
4. Power up the UPS by closing the UPS AC input. Verify that the UPS status light is on and that no alarm lights are on. Record this verification.
5. If this is the first time booting the UPS, then using the DMM, measure the AC output voltage of the UPS at the UPS return to rack plug on the shelter wall. The measured voltage shall be within the range of 108 to 132 volts. Record the measured DMM voltage.
6. After verifying that all rack equipment power switches are in the OFF position, reconnect the rack power cable to the UPS rack power socket at the shelter wall.
7. Turn on the RSC using the switch on the front panel
8. Turn the monitor power switch on and verify that the monitor is on by observing that the green power LED is illuminated.
9. Switch the monitor to the MIU. Turn the MIU power switch to on, verify that the MIU is on (power LED is illuminated) and that the fans are pulling air in. Login as "tlstech" with the login password provided. Test the mouse by verifying that the pointer moves and that the right and left mouse buttons work.
10. Switch the monitor to Base2. Turn Base2 power switch to on, verify that the Base2 is on (green power LED is illuminated) and that the fans are pulling air in. Login as "tlstech" with the login password provided. Test the mouse by verifying that the pointer moves and that the right and left mouse buttons work.
11. Switch the monitor to Base1. Turn Base1 power switch to on, verify that the Base1 is on (green power LED is illuminated) and that the fans are pulling air in. Login with the login name and password provided.
12. Turn the GTU power switch to on, verify that the GTU is running by observing that the red LED on the power switch is on and that the fan is pulling air in.
13. Turn the Interrogation transmitter power switch to on; confirm that the Interrogation transmitter is running by verifying that the fans are pulling air into the transmitter and that the green LED next to the power switch is illuminated.
14. Close the uplink ob-light breaker and verify that the ob-light is illuminated.
15. Connect the UPS CAT 5 network cable from the UPS network card to the network hub.
16. Record verification of proper power up of Shelter equipment.

4.9 Sensor Power On

1. Power on the ASA sensor by closing the circuit breaker at the base shelter labeled "ASA". Wait 2 minutes to ensure that the sensor has sufficient time to boot.
2. On the Base1 CPU, type "ping ASA". If the sensor does NOT respond as an ASA, verify that the AOA is up and running by typing "ping ESA", and verifying that the sensor replies. Type "Ctrl+c" to stop the ping test. If no response, troubleshoot the sensor network connections. If it replies as an ESA then adjust the boot software on the sensor.
3. Power on the ESA sensor by closing the circuit breaker at the base shelter labeled "ESA". Wait 2 minutes to ensure that the sensor has sufficient time to boot. Verify that it is up and running by typing "ping ESA". Type "Ctrl+c" to stop the ping test.
4. Verify that the Obstruction Light at both the ESA and ASA locations are operational.
5. Record verification of the successful sensor ping tests, and that obstruction lights are illuminated.

4.9.1 Sensor 920-00400 Configuration as ASA or ESA

This model number sensor has a physical manual switch located in the sensor enclosure to configure the sensor as an ESA or and ASA. To configure the sensor as an ASA:

1. Power off the ESA sensor at the breaker panel inside the TLS shelter.

2. Go to the ASA location and open the sensor door, identify the switch labeled “Configuration” and set the switch to “ASA”
3. Close and secure the sensor door and power on the ASA.

To configure the sensor as an ESA:

1. Power off the ESA sensor at the breaker panel inside the TLS shelter.
2. Go to the ESA location and open the sensor door, identify the switch labeled “Configuration” and set the switch to “ESA”
3. Close and secure the sensor door and power on the ESA.

4.9.2 Sensor 920-00256 Configuration as ASA or ESA

TLS sensors with subject part number can be configured as either an ASA or ESA. Examine the part number on the AOA sensor provided with the TLS kit to determine the applicability of this procedure. Spare sensors are configured as an ESA. Only the sensor that must be reconfigured should be powered on during the procedure.

To configure a sensor as an ASA, perform the following steps:

1. Power off the ESA sensor location.
2. Power off the ASA sensor location.
3. Install the spare sensor in the ASA location.
4. Power on the ASA sensor location.
5. From Base1, type "arp -d esa".
6. From Base1 remote login to the sensor by typing "rlogin esa".
7. Type "cd /boot/images".
8. In the ESA shell, type “./setup_laoa”. When prompted “reboot now?” type “y” and press the return key. The sensor will reboot.
9. Re-boot the Base1 CPU (to clear the network nodes registered).
10. After the sensor and Base1 have rebooted, verify that the sensor is now an ASA by typing "ping ASA" and checking that the sensor replies successfully. Type "Ctrl+c" to stop the ping test.
11. Power on the ESA sensor location.
12. From Base1, verify that the ESA is up by typing "ping ESA" and checking that the sensor replies successfully. Type "Ctrl+c" to stop the ping test.

Configure a sensor as an ESA by performing the following steps:

1. Power off the ESA sensor location.
2. Power off the ASA sensor location.
3. Install the spare sensor in the ESA location.
4. Power on the ESA sensor location.
5. From Base1, type "arp -d esa".
6. From Base1 remote login to the sensor by typing "rlogin asa".
7. Type "cd /boot/images".
8. In the ASA shell, type “./setup_gaoa”. When prompted “reboot now?”, type “y” and press the return key. The sensor will reboot.
9. Re-boot the Base1 CPU (to clear the network nodes registered).
10. After the sensor and Base1 have rebooted, verify that the sensor is now an ESA by typing "ping ESA" and checking that the sensor replies successfully. Type "Ctrl+c" to stop the ping test.

11. Power on the ASA sensor location.
12. From Base1, verify that the ASA is up by typing "ping ASA" and checking that the sensor replies successfully. Type "Ctrl+c" to stop the ping test

4.9.3 Antenna Heater Controller Deicer On (Option)

Set the control switch on the Antenna Heater Controller display panel to manual.

After 15 minutes, go to each antenna and ensure that each is warm to the touch.

Record verification that the Antenna Heater Controller is working.

Set the control switch to off.

4.10 Cal/BIT Power On

1. Power on the Cal/BIT by closing the circuit breaker at the Shelter labeled "Cal/BIT". Verify that the Obstruction light at the Cal/BIT location is operational.
2. Record verification that the Cal/BIT obstruction lights are illuminated.

4.11 UPS Battery Operational Check

1. To initiate a battery test, press the ON button for at least one second while operating from utility power with no alarm conditions present.
2. If only three of the five Battery LEDs illuminate, allow the UPS to recharge the batteries for 24 hours.
3. After 24 hours, retest the batteries.
4. After the batteries have been retested, if only three of the five Battery LEDs illuminate, contact ANPC or authorized local partner for service information.
5. Record verification that the battery self test completed successfully.

4.11.1 UPS Status Indicators: Backup Mode

When the system has gone into UPS Backup mode it is operating on battery power. The operator will see the UPS Status line reading ON BATTERY, with the minutes to shutdown provided.



Figure 4-3 Maintenance Interface UPS Status

4.12 Remote Units Power On

4.12.1 RCU Power On

1. Power on and login as "RCU Operator" with the login provided password. Verify that the keyboard, monitor and mouse are working properly. The RCU application will start at boot.
2. From the RCU, open a "Command Prompt" shell, type "ping base1" and press the "Enter" key. Verify that replies from base1 occur. Type "ping base2" and press the "Enter" key. Verify that replies from base2 occur. Close the "Command Prompt" shell.
3. On the RCU, start the RCU application and verify that the system mode display shows the current system mode. Base software is not running yet, so "NO COMM" will be displayed at this time.
4. Record verification that the RCU link is established.

4.13 System CPU Clock Synchronization

1. On the MIU, verify that the time zone, date and time are correctly set for the site location. If incorrect, use the "Adjust Date/Time" function of the Windows operating system (or double - click on the time display in the lower right corner) to select local time zone, date and time. Save these settings.
2. Switch to Base2 and repeat the above step. If date and / or time zone are incorrect, use the "Adjust Date/Time" function of the Windows operating system to set the local time zone, and date. Save these settings.
3. Switch to Base1 and at the command prompt type the following (without quotes), "date" and hit the "enter" key to display the current time zone, date and time. Check the time zone, which will be displayed as three capital letters (e.g. "PST" for Pacific Standard Time). If the time zone is incorrect, contact ANPC for directions on setting the time zone file (etc/config/sysinit.1) for Base1.
4. On the MIU, start the SyncClock Tool, located in the c:\Tools folder.
5. Verify that the correct time and date is displayed in the SyncClock window. Check the radio button labeled "x86" and uncheck the checkboxes labeled "gaoa" and "laoa". Press the "Sync" button. Click on the "Close" button to exit the SyncClock application.
6. Wait one minute to allow CPU clocks to refresh.
7. Note the current time and date on the MIU, then check that the times and dates match this on Base1 and Base2.
8. Record validation that the Shelter CPU clocks are synchronized.

4.14 System Software Startup

1. On the MIU, open the MI application.
2. From the MI, select "System Status", and then "Network Test". Commence the communication check, and verify that each of the Shelter computers and sensors respond. Exit the Network Test dialog, and system status.

The screenshot shows the 'System Settings' window with the following sections and values:

- Diagnostic Mode BIT Schedule:**
 - Retry Interval: 10 seconds
 - Retries: 0
 - Daily BIT Time: 12:00:00 AM
- Maintenance Mode BIT Schedule:**
 - Interval: 5 minutes
- GTU Track BIT Schedule:**
 - Interval: 60 minutes
- F1 Cal/BIT Position:**
 - Near: 5 μ s
- UPS Shutdown:**
 - 10 minutes
- Environment:**
 - Barometric Pressure: 29.92 in Hg
 - Transition Level: 18000 feet
 - Max Noise Level: 40 counts
 - ☐ Compress Sensor Data
- GTU Type:**
 - ☒ 920-00101 ☐ 920-00402
- Service Volume:**
 - Az Volume Edges: $\pm$ 35 degrees
 - EI Volume Lower Edge: 0.6 degrees
 - EI Volume Upper Edge: 7 degrees
 - Localizer Range: 18 NM
 - Glide slope Range: 12 NM
 - EI Az Volume Edges: $\pm$ 10 degrees
- Acquire Settings:**
 - Acquire Timeout: 2 minutes
 - Reacquire Limit: 0
- Interrogation:**
 - Guidance: 10 Hz
 - Surveillance: 2 Hz
 - ☒ Enable P2
- TOA Delay Settings:**
 - ESA: 0 ns
 - ASA: 0 ns
 - ATA: 0 ns
- Data Recording:**
 - Record Track RAM Files: ☒ Off ☐ Guidance ☐ All
 - ☐ Record BIT RAM Files
 - ☒ Record Raw Files
 - ☐ Record GTU Files
 - ☒ Record Track BIT Files
 - ☒ Log Track Info Messages
 - ☒ Log DQM Messages
 - First Character of Raw File Name: x
 - Send FTM Data for this RCU: RCU1
- Rack Config:**
 - ☒ Single ☐ Rack A ☐ Rack B

Buttons at the bottom: OK, Cancel, Apply.

Figure 4-4 System Settings Window

3. Click on the "SYSTEM SETUP" button on the Maintenance Interface. Then click on the "SETTINGS" button. Verify the following default settings and change as needed:
 - a. Diagnostic Mode BIT Schedule:
 - <Retry Interval 10 seconds>
 - <Retries 2>
 - <Daily BIT Time: <12:00:00AM>
 - b. Maintenance Mode BIT Schedule:
 - <5 minutes>
 - c. GTU Track BIT Schedule:
 - <60 minutes>
 - d. F1 Cal/BIT Position
 - <Near: 5 μ s>
 - e. UPS Shutdown:
 - <10 minutes>
 - f. Service Volume:
 - <Localizer Azimuth Volume Edge: $\pm$ 40degrees>
 - <Elevation Volume Lower Edge: 0.9 degrees>
 - <Elevation Volume Upper Edge: 7 degrees>
 - <localizer Range: 25 NM>
 - <Glide slope Azimuth Volume Edge: 12 degrees>
 - <Glide slope Range: 12NM>
 - g. Interrogation:

- <Guidance: 20 Hz>
 - <Surveillance: 2 Hz>
 - <Enable P2: Checked> (if a P2 suppression pulse is installed)
- h. ESA Tilt Sensors:
- <Enabled>
- i. Diversity Antenna:
- <Enabled>
- j. Data Recording:
- <Record Track RAM Files: Guidance>
 - <Record BIT RAM Files: Not Checked>
 - <Record Raw Files: Checked>
 - <Record GTU Files: Checked>
 - <Record Track BIT Files: Checked>
 - <Log Track Info Messages: Checked>
 - <Log DQM Messages: Checked>
 - <First Character of Raw File Name: x>
 - <Send FTM Data for this RCU: RCU1>
 - <Rack ID: NONE, A or B> select appropriate designator
4. In the "System Settings" window click on the "OK" button.
 5. On the MI, start the system software. Observe that the System Mode indicator goes from "NO COMM WITH BASE SW", to "INITIALIZING", to "DIAGNOSTIC", then after about 15 seconds it will transition to "MAINTENANCE" or "TRACK".
 6. From the Maintenance Interface click on "Help" on the pull-down menu, then "System Information". Record the successful start of system software and the version # for the software.

4.15 GTU Setup

This procedure describes the software calibration of each GTU. Calibration of the GTUs is performed using the MI GTU Test tool to transmit selected DDM values, using a PIR to measure transmitted DDM values, and using the GtuTxCal tool to enter measured DDM values into the data files. This process is repeated for each GTU installed in the system.

4.15.1 Tools and Materials Required

The following is required to perform this procedure.

- Portable ILS Receiver (PIR)
- Two handheld radios.
- Full system hardware with a GTU installed.
- Two operators (one to operate the MIU and one to operate the PIR).
- Activity sheet (included with this procedure).

4.15.2 Special Instructions

Coordinate a TLS service outage with appropriate airfield personnel who will notify ATC and other local and federal agencies as required.

1. Verify that the PIR has a valid / current calibration sticker prior to using.
2. The system software uses a single set of GTU transmit calibration settings for the main runway.

3. The sign of DDM values indicated on the PIR is opposite that indicated in the GTU Test tool. The GTU transmit calibration uses the sign indicated on the PIR for values measured and entered in the GtuTxCal tool.
4. In the GTU Test tool, the **Use transmit calibration** checkbox must be unchecked when performing the transmit calibration, and checked when verifying the transmit calibration and performing a monitor calibration.
5. Two copies of the activity sheet included in this procedure shall be printed for each GTU being calibrated. One sheet shall be filled in by the MIU operator and one shall be filled in by the PIR operator. Data from both sheets shall be cross-checked by both operators.

4.15.2.1 GTU Setup Procedure

Performing a calibration of the GTU transmitter consists of the following steps:

1. Setting up the PIR.
2. Measuring transmit power with a PIR.
3. Measuring transmit DDM values with a PIR.
4. Entering transmit DDM values using the GtuTxCal tool.
5. Performing a final monitor calibration.
6. Verifying the transmit calibration.
7. Verifying the monitor calibration.

Listed below are the procedures necessary to complete the above steps:

1. Setup the PIR: Place the PIR on the main runway side, approximately 30 feet in front of the uplink tower. When measuring GTU transmitter power, the PIR should be placed (or held) 4 to 5 feet above the ground. When measuring DDM values, the PIR can be placed on the ground. Note that the PIR antennas shall be extended for localizer measurements and reduced for glide slope measurements.
2. Measure GTU Power and SDM: Stop the system software from the MI. Open the GTU Test tool on the MI. In the GTU Test dialog, select the tab corresponding to the GTU being calibrated. Uncheck the checkbox labeled Use transmit calibration. Turn on the GTU transmitters by pressing the GS on button.
3. Localizer Power: Have the PIR operator configure the PIR to measure localizer power. The antennas on the PIR must be extended for localizer measurements. Have the PIR operator hold the PIR about 4 to 5 feet above ground and measure the localizer power. The received power shall be -5 ± 5 dBm (i.e., -10 to 0 dBm). Record the localizer power. Configure the PIR to measure localizer SDM and record this value.
4. Glide slope Power: Have the PIR operator configure the PIR to measure glide slope power. The antennas on the PIR must be collapsed for glide slope measurements. Have the PIR operator hold the PIR about 4 to 5 feet above ground and measure the glide slope power. The received power shall be -15 ± 5 dBm (i.e., -20 to -10 dBm). Record the power on the activity sheet. Configure the PIR to measure glide slope SDM and record this value on the activity sheet.
5. Measure Transmit DDM Values: Stop the system software from the MI. Open the GTU Test tool on the MI. In the GTU Test dialog, select the tab corresponding to the GTU being

- calibrated. Uncheck the checkbox labeled Use transmit calibration. Turn on the GTU transmitters by pressing the GS On button.
6. Have the PIR operator configure the PIR to measure localizer DDM.
 7. For each item in the localizer DDM popup menu in the GTU Test dialog, perform the following steps (starting with the first item in the popup menu):
 - a. Select the DDM value.
 - b. Wait 2 seconds.
 8. Tell the PIR operator the value of the current DDM selection and have the PIR operator report the measured DDM value (with the sign as indicated on the PIR). Record the measured DDM on the activity sheets of both operators.
 9. Have the PIR operator configure the PIR to measure glide slope DDM.
 10. For each item in the glide slope DDM popup menu in the GTU Test dialog, perform the following steps (starting with the first item in the popup menu):
 - a. Select the DDM value.
 - b. Wait 2 seconds.
 11. Tell the PIR operator the value of the current DDM selection and have the PIR operator report the measured DDM value (with the sign as indicated on the PIR). Record the measured DDM on the activity sheets of both operators.
 12. Close the GTU Test dialog and exit the MI application.
 13. Compare Activity Sheet DDM Values: Compare the DDM values recorded by the MIU operator and the GTU operator. Resolve any discrepancies.
 14. Enter Transmit DDM Values: Exit the MI application if it is running. Open the GtuTxCal tool, located in the C:\Tools directory.
 15. Press the Load... button in the GtuTxCal tool and open settings.dat in the MIU C:\Run directory. Select the tab corresponding to the GTU being calibrated. Press the Default button. Enter DDM values from the into the edit fields of the GtuTxCal tool. One operator shall enter the values and the other operator shall verify the entries.
 16. Press the Save... button and save to settings.dat in the MIU C:\Run directory. A message box will be displayed asking if you want to save the file to the bases. Press the Yes button.
 17. Exit the GtuTxCal tool.
 18. Perform a Final GTU Monitor Calibration: Once the GTU transmitter is calibrated, it is necessary to recalibrate the GTU monitor. Use the GTU Test tool and enter the SDM values recorded by the PIR to perform the monitor calibration.
 19. Verify GTU Transmit Calibration: Verifying the GTU transmit calibration is essentially the same steps as recording DDM values above, with the important exception that the checkbox labeled "Use transmit calibration" must be checked in the GTU Test dialog. Record all DDM values to the verification table of the activity sheet.
 20. Verify GTU Monitor Calibration: Perform a final GTU monitor calibration using the GTU Test tool.
 21. Archive Activity Sheet: Scan the Activity Sheets to a PDF file and save them on the server.

4.15.2.2 Uplink Monitor Configuration

Complete the Preventive Maintenance procedure GTU Check at this time.

Table 4-2 GTU Transmit Calibration Activity Sheet

Date and Time		
Airfield Code		
GTU number, Serial Number and frequency	Example: 1 / 8231 / 108.35	
Operators		
LOC DSA / Atten	Example: +10.0 / 2.42V	
Localizer Power		
GS DA / Atten		
Glide Slope Power		
Localizer SDM		
Glide Slope SDM		
Calibration		
Commanded DDM from GTU Test Tool	Measured DDM from PIR	
	Localizer	Glide Slope
+0.24	N/A	
+0.22	N/A	
+0.20		
+0.18		
+0.15		
+0.12		
+0.09		
+0.06		
+0.03		
0.0		
-0.03		
-0.06		
-0.09		
-0.12		
-0.15		
-0.18		
-0.20		
-0.22	N/A	
-0.24	N/A	

Verification: GTU _____				
Commanded DDM from GTU Test Tool	Measured DDM from PIR			
	Main Runway		Reciprocal Runway	
	Localizer	Glide Slope	Localizer	Glide Slope
+0.24	N/A		N/A	
+0.22	N/A		N/A	
+0.20				
+0.18				
+0.15				
+0.12				
+0.09				
+0.06				
+0.03				
0.0				
-0.03				
-0.06				
-0.09				
-0.12				
-0.15				
-0.18				
-0.20				
-0.22	N/A		N/A	
-0.24	N/A		N/A	

4.16 Input Survey Data

- At the Maintenance Interface main window, click on "SYSTEM SETUP" button. At the System Setup main window, click on the "Survey" button.
- Use survey data sheet to complete the following data entry:
 - Input the range and azimuth Angle for the ASA Reference Antenna; ATA; Cal/BIT Antenna; and Interrogation Antenna.
 - Input the ESA reference antenna range, azimuth angle, and elevation angle. Use 44 inches for "tower base plate separation from reference phase center".
 - Input the ESA separation from reference antenna for the Low, Medium and High antennas.
 - Input the tower base plate separation from the reference antenna phase center.
 - Input the MSL altitude and runway heading.
- Press "OK" to close the Survey data. Press "Close" to exit "System Setup" menu and return to the MI main menu.
- Record verification that survey data was entered.

4.17 Tilt Sensor Configuration (option)

The tilt sensor configuration serves the purpose of monitoring the TLS ESA tower by providing a measurement of tower tilt to the monitor. If tilt sensors are installed, perform the following data check:

1. Using Notepad on Base2, open the "c:\run\tilt.dat" file.
2. Verify that the data in the tilt.dat file correctly matches the serial numbers and location (perpendicular or parallel) for the installation. Also verify that the table matches tilt sensor serial numbers listed on the tilt sensors.
3. If the data do not match, contact ANPC for a properly formatted replacement file. Install the new file on CPU1, CPU2 and the MIU.
4. Record validation of tilt data.

The following steps verify correct tilt sensor cable connections and establish final leveling of the tilt sensors using BIT data.

5. Using Notepad on Base2, open the file "c:\run\bit\esa.bit". Note the data for the parallel and perpendicular tilt sensors.
6. Physically remove the connector at the perpendicular tilt sensor at the ESA.
7. On the Maintenance Interface (Test Mode), complete three BIT cycles.
8. Using Notepad on Base2, re-open the file "c:\run\bit\esa.bit". Verify that the data for the perpendicular tilt sensors has changed to -1.4 degrees (full-scale no signal indication) and is the same for both *tilt measurement* and *tilt test*. Record this verification.
9. Re-connect the perpendicular tilt sensor connector.
10. Examine the BIT data. Determine which way to adjust the tilt sensor bracket to achieve measurements within ± 0.1 degrees. The sign convention for tilt sensor angle measurement follows the left-hand rule; with the hand over the tilt sensor and thumb aligned with the cable connector, the fingers will point in the direction of positive angle measurements.
11. Adjust the tilt sensors to be level. This is an iterative process that requires one person at the ESA tower to physically adjust the tilt bracket and another person at the Shelter to observe and report on the BIT cycle measurement of tilt between each change.

NOTE: The tilt sensors mounting bracket will require slight adjustments between running BIT cycles to achieve the final alignment.

12. On the Maintenance Interface (Test Mode), run continuous BIT and display the sensor data window for the ESA.
13. Using MultiBIT re-open the file "c:\run\bit\esa.bit", examine the last 5 data points and determine the average of the perpendicular and parallel tilt data. Enter the average measurement for the perpendicular sensor on the monitor limits -> ESA limits -> tilt sensor tab. Save data to base CPUs. Record final value on the TRR.
14. When the process is complete, return to the MI main window.

4.18 AOA Sensor, Interrogation Transmitter, Guidance Transmitter Monitor Limits

This procedure configures initial monitor limits.

1. Start the Test Mode software by selecting Tools on the pull-down menu and then Test Mode.
2. Command BIT to run at least 10 times, and then stop the system software.

- Using MultiBIT data tool, open \\Base2\C\Run\BIT\uplink_1.bit. Review data, select tools/compute nominal values and save.

The system will verify that the file was copied to CPU 2 and ask whether to send the file to CPU 1. Answer "Yes" when prompted to save the file to CPU 1.

- Using MultiBIT data tool, open \\Base2\C\Run\BIT\ata.bit. Review data, select tools/compute nominal values and save.

The system will verify that the file was copied to CPU 2 and ask whether to send the file to CPU 1. Answer "Yes" when prompted to save the file to CPU 1.

- Using MultiBIT data tool, open \\Base2\C\Run\BIT\esa.bit. Review data, select tools/compute nominal values.

The system will verify that the file was copied to CPU 2 and ask whether to send the file to CPU 1. Answer "Yes" when prompted to save the file to CPU 1.



Parameter	Current	Computed	Delta	Std Dev	Units	Save
TOA Near	6421.9	6422.0	0.1	0.000	ns	☒
TOA Far	396421.9	396422.0	0.1	0.000	ns	☒
Amp Ref	1650	1648	-2	0.000	mV	☒
Amp Low	1474	1468	-6	0.000	mV	☒
Amp Medium	1509	1507	-2	0.000	mV	☒
Amp High	1420	1414	-6	0.000	mV	☒
Phase L	4	4	0	0.000	counts	☒
Phase M	88	88	0	0.000	counts	☒
Phase H	254	254	0	0.000	counts	☒
Frequency	1090.000	1090.018	0.018	0.000	MHz	☐
Tilt Para	0.000	0.000	0.000	0.000	degrees	☐
Tilt Perp	0.000	0.000	0.000	0.000	degrees	☐

Data Path: C:\DTLS\dtls_tls\run\base2

☐ Show Current Limits ☐ Show New Limits

Less << Save ... Close

Figure 4-5 Compute Nominal Values

- Using MultiBIT tool, open \\Base2\C\Run\BIT\asa.bit. Review data, select tools/compute nominal values and save.

The system will verify that the file was copied to CPU 2 and ask whether to send the file to CPU 1. Answer "Yes" when prompted to save the file to CPU 1.

- Using MultiBIT data tool, select "Open base1 interrog.bit" in the File menu. Review data, select tools/compute nominal values and save.

The system will verify that the file was copied to CPU 2 and ask whether to send the file to CPU 1. Answer "Yes" when prompted to save the file to CPU 1.

- On the MI, press the "Archive..." button, and then click "Run". When the process is complete, click "Close" to return to the MI main window.
- From the MI, start system software. Wait for the system to complete "Diagnostic" mode and enter "Track" mode.

10. Enter Test mode. From Test Mode, command BIT at least 5 more times and exit Test Mode to MI main dialog.
11. On the MI, go to the System Status Window, select "Error History" and wait for the screen to update. Verify that all of the LRU status indicators are green. Exit to MI main screen.

4.19 Initial Cal/BIT TOA and Frequency Settings

Before the TOA delay settings can be calculated, the survey data must be entered and the system must be passing BIT.

4.19.1 TOA F1 Tuning

The TLS AOA sensor performance is monitored using the BIT signal transmitted by the Cal/BIT unit. The time position where that BIT pulse arrives in the sensor buffer should be optimized to achieve the lowest measurement jitter using the following steps.

- "T_I" refers to the optimal initial F1 Cal/Bit Position and is usually 5.2 μ S. If procedure is being done as part of a replacement of an LRU, "T_I" refers to the current nominal setting for F1 Cal/BIT Delay.
 - Use both the F1 delay on the setting dialog of the MI and the TOA monitor nominal value of each sensor (ASA / ESA) to adjust the CAL/BIT pulse until it is well centered within the bit ram window.
1. On the MI, press the "Archive..." button. In the Archive dialog, press the Run button. When the process is complete, click "Close" to return to the MI main window.
 2. On the MI, in "Settings", enable BIT-RAM, Set the F1 delay to T_I.
 3. On the MI, start the system software. Enter and exit Test mode, and then stop system software.
 4. In "Settings", set the F1 delay to (T_I + .05 μ S).
 5. On the MI, start the system software. Enter and exit Test mode, and then stop system software.
 6. On the MI, in "Settings", set the F1 delay to (T_I + .10 μ S).
 7. On the MI, start the system software. Enter and exit Test mode, and then stop system software.
 8. On the MI, in "Settings", set the F1 delay to (T_I + .15 μ S).
 9. On the MI, start the system software. Enter and exit Test mode, and then stop system software.
 10. On the MI, in "Settings", set the F1 delay to (T_I + .20 μ S).
 11. On the MI, start the system software. Enter and exit Test mode, and then stop system software.
 12. On the MI, in "Settings", set the F1 delay to (T_I + .25 μ S).
 13. On the MI, start the system software. Enter and exit Test mode, and then stop system software.
 14. Open the MultiBIT tool. Open all ASA BIT-RAM files (asa*b*.ram) in the [\\base2\c\run\raw](#) directory.
 15. In MultiBIT, examine the TOA and TOA jitter. Enable the Cal/BIT limits display and compare the measurement data to the monitor limits. Determine the F1 setting that resulted in the lowest TOA jitter.
 16. In MultiBIT, open all ESA BIT-RAM files (esa*b*.ram) in the [\\base2\c\run\raw](#) directory.
 17. In MultiBIT, examine the TOA and TOA jitter. Enable the Cal/BIT limits display and compare the measurement data to the monitor limits. Determine the F1 setting that resulted in the lowest TOA jitter.

18. Determine the F1 setting that resulted in the lowest TOA jitter for both ESA and ASA. The TOA jitter should be between 2 to 4 ns. Record this best F1 setting.
19. On the MI, in "Settings", set the F1 Cal/BIT Position to the optimal setting (as determined in the previous step) and disable BIT RAM recording.
20. On the MI, press the "Archive..." button. In the Archive dialog, press the Run button. When the archive finishes, press the Close button.
21. On the MI, start the system software.
22. On the MI, enter Test mode.
23. In the Test Mode dialog, set BIT to run continuously after 5 cycles disable continuous BIT.
24. Exit Test mode. Stop the TLS software, exit the MI.
25. Use the multibit tool to set monitor limits values for the ATA, ASA and ESA.
26. On the MI, start the system software.

4.19.2 Cal/Bit Power Setting

Use the following steps to bring the Cal/BIT level within the measurement limits of the TLS AOA sensors.

1. Examine the Cal/BIT amplitude for all three channels of the ASA sensor. Use the MultiBIT data tool to open the file \\Base2\C\Run\bit\asa.bit.
2. Repeat the previous step for the ESA and ATA.
 - Ensure that power is removed from the Cal/BIT transmitter before disconnecting RF and adjusting the attenuator. There should be approximately 75 mV of change for each 3 dB change in attenuation.
3. If any of the channels for either sensor have Cal/BIT amplitude above 1950 mV, then decrease the Cal/BIT power by adding attenuation at the SMA output connector of the RF module inside the Cal/BIT transmitter in 3 dB steps (3dB and 6dB pads are shipped with the Cal/BIT transmitter taped to the inside of the cover). If the lowest of the amplitude measurements is below 1500 mV, then remove attenuation inside the Cal/BIT transmitter.
 - Adjusting the antenna alignment of the Cal/BIT may be beneficial for optimizing the Cal/BIT power.
4. Enter final Cal/BIT received amplitudes into the corresponding locations within the Monitor Limits settings interface for both the ASA and ESA. Press the OK button to save changes. Record final attenuation inserted in the Cal/BIT path.

4.20 Setting System Monitor Values

This procedure is designed to set the monitor limits for each of the system's functional assemblies. This includes the three sensors, the interrogation transmitter and each GTU. Each has to be set before the system can be put into an operational mode. The MultiBIT tool is used to automatically calculate the monitor limits..

If the system has been operating and the RAM files from the previous day have not been archived and cleaned, do so at this time before configuring the Cal/BIT monitor nominal values.

Command BIT Procedure:

- a. On the MI click **Start** to start the system software.

- b. After the system transitions to either Track or Maintenance mode, click **Test Mode** in the **System** menu to put the system into the Test mode of operation.
- c. set BIT to run continuously and after 5 iterations of BIT, stop continuous BIT mode.
- d. Exit the Test mode dialog by clicking Close.
- e. Click **Stop** to stop the system software.

4.20.1 ESA, ASA and ATA Monitor Values

This procedure sets the monitor values for the ESA, ASA and ATA sensor assemblies. This procedure requires first performing the Command BIT procedure described in Section 4.20 .

- a. In the MI, click **Review BIT Monitor Data (MultiBIT)** in the **Tools** menu to open the MultiBIT tool.
- b. In MultiBIT, click **Open ESA BIT** in the **File** menu.
- c. In MultiBIT, click **Compute Nominals** in the **Tools** menu. The tool will display the current settings and the computed values. Color coding is used to show where current settings are beyond limits. Red indicates the current setting is beyond alarm limits and yellow indicates it is beyond alert limits. Figure 4-6

Computed Nominals						
Parameter	Current	Computed	Delta	Std Dev	Units	Save
TOA Near	6421.9	6421.0	-0.9	0.000	ns	☒
TOA Far	396421.9	396421.0	-0.9	0.000	ns	☒
Amp Ref	1650	1414	-236	0.000	mV	☒
Amp Low	1474	1234	-240	0.000	mV	☒
Amp Medium	1509	1390	-119	0.000	mV	☒
Amp High	1420	1179	-241	0.000	mV	☒
Phase L	0	0	0	0.000	counts	☒
Phase M	0	0	0	0.000	counts	☒
Phase H	0	0	0	0.000	counts	☒
Delta Phase L	0	0	0	0.000	counts	☒
Delta Phase M	0	0	0	0.000	counts	☒
Delta Phase H	0	0	0	0.000	counts	☒
Frequency	1090.000	1090.018	0.018	0.000	MHz	☐
Tilt Para	0.000	0.000	0.000	0.000	degrees	☐
Tilt Perp	0.000	0.000	0.000	0.000	degrees	☐

Data Path: \\base2\c\run

☐ Show Current Limits ☐ Show New Limits

Less << Save Close

Figure 4-6 ESA Compute Nominals

- d. In MultiBIT click **Save** [Figure 4-6]. This will save the monitor limits to both bases.
- e. In MultiBIT, click **Open ASA BIT** in the **File** menu.

- f. In MultiBIT, click **Compute Nominals** in the **Tools** menu. The tool will display the current settings and the computed values. Color coding is used to show where current settings are beyond limits. Red indicates the current setting is beyond alarm limits and yellow indicates it is beyond alert limits [Figure 4-7].

Computed Nominals						
Parameter	Current	Computed	Delta	Std Dev	Units	Save
TOA Near	5746.7	5747.0	0.3	0.000	ns	☒
TOA Far	395746.7	395747.0	0.3	0.000	ns	☒
Amp Ref	1750	1515	-235	0.000	mV	☒
Amp Low	1744	1507	-237	0.000	mV	☒
Amp Medium						
Amp High	1620	1484	-136	0.000	mV	☒
Phase L	0	0	0	0.000	counts	☒
Phase M						
Phase H	0	0	0	0.000	counts	☒
Delta Phase L	0	0	0	0.000	counts	☒
Delta Phase M						
Delta Phase H	0	0	0	0.000	counts	☒
Frequency	1090.000	1090.018	0.018	0.000	MHz	☐
Tilt Para						
Tilt Perp						
Data Path: \\base2\c\run						
☐ Show Current Limits ☐ Show New Limits						
Less << Save Close						

Figure 4-7 ASA Compute Nominals

- g. In MultiBIT click **Save** [Figure 4-7]. This will save the monitor limits to both bases.
- h. In MultiBIT, click **Open ATA BIT** in the **File** menu.
- i. In MultiBIT, click **Compute Nominals** in the **Tools** menu. The tool will display the current settings and the computed values. Color coding is used to show where current settings are beyond limits. Red indicates the current setting is beyond alarm limits and yellow indicates it is beyond alert limits [Figure 4-8].

Computed Nominals						
Parameter	Current	Computed	Delta	Std Dev	Units	Save
TOA Near	6579.7	6576.0	-3.7	0.000	ns	☒
TOA Far	396579.7	396576.0	-3.7	0.000	ns	☒
Amp Ref						
Amp Low						
Amp Medium	1546	1304	-242	0.000	mV	☒
Amp High						
Phase L						
Phase M						
Phase H						
Delta Phase L						
Delta Phase M						
Delta Phase H						
Frequency						
Tilt Para						
Tilt Perp						

Data Path: \\base2\c\run

☐ Show Current Limits ☐ Show New Limits

Less << Save Close

Figure 4-8 Azimuth Time of Arrival (ATA) Compute Nominals

- j. In MultiBIT click **Save** [Figure 4-8]. This will save the monitor limits to both bases.
- k. In MultiBIT, click **Exit** in the **File** menu to exit the application.

4.20.2 GTU Nominal Values

- a. In MultiBIT, click **Open Uplink_1 BIT** in the **File** menu.
- b. In MultiBIT, click **Compute Nominals** in the **Tools** menu. The tool will display the current settings and the computed values. Color coding is used to show where current settings are beyond limits. Red indicates the current setting is beyond alarm limits and yellow indicates it is beyond alert limits [Figure 4-8].
- c. Repeat steps a. and b. for Uplink_2, Uplink_3 and Uplink_4 if they exist (up to four GTUs may be installed)

4.20.3 Interrogation Transmitter Nominal Values

This procedure sets the monitor values for the interrogator assembly. This procedure requires first performing the Command BIT procedure described in Section 4.20 .

- a. In the MI, click **Review BIT Monitor Data (MultiBIT)** in the **Tools** menu to open the MultiBIT tool.
- b. In MultiBIT, click **Open Interrogator BIT** in the **File** menu [Figure 4-9].

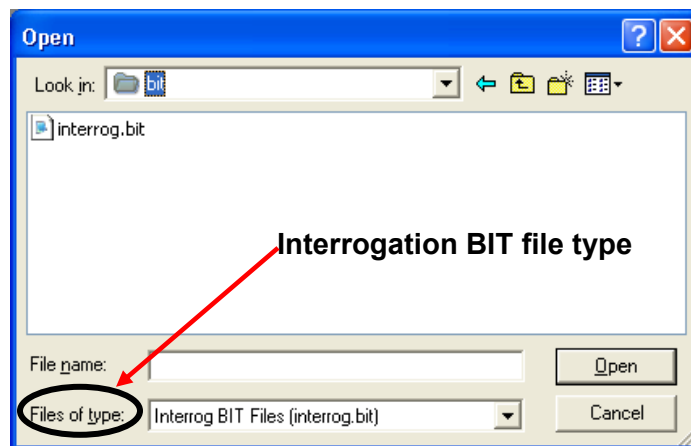
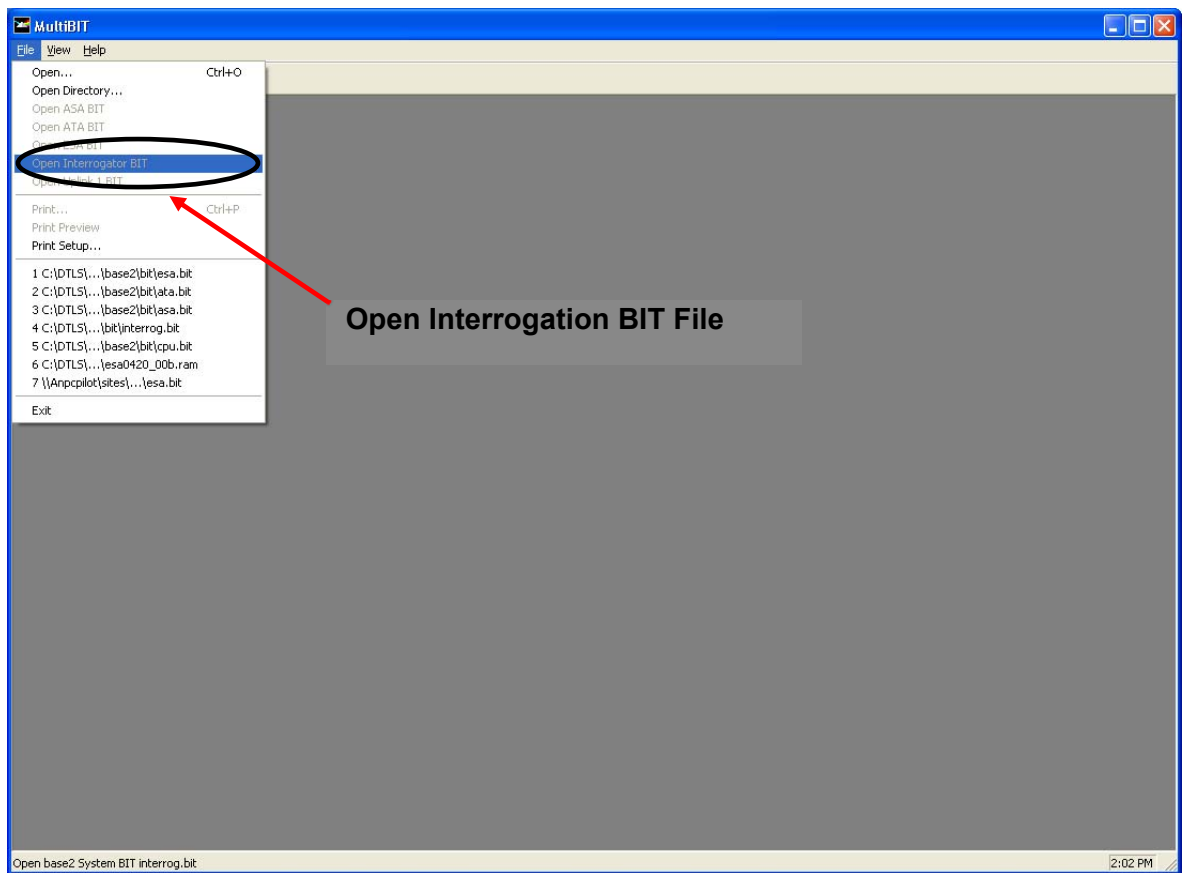


Figure 4-9 Interrogation Transmitter Open BIT File

- c. In MultiBIT, click **Compute Nominals** in the **Tools** menu. The tool will display the current settings and the computed values. Color coding is used to show where current settings are beyond limits. Red indicates the current setting is beyond alarm limits and yellow indicates it is beyond alert limits [Figure 4-10].

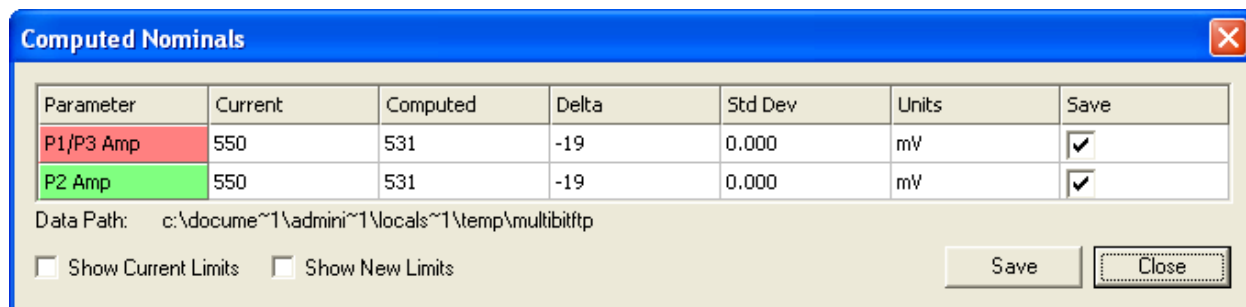


Figure 4-10 Interrogation Transmitter Compute Nominal Values

- d. In MultiBIT click **Save** [Figure 4-10]. This will save the monitor limits to both bases.
- e. In MultiBIT, click **Exit** in the **File** menu to exit the application.

4.20.4 Validating Nominal Monitor Values

This procedure is designed to help the technician validate that the monitor values that have been set are correct and will allow the system to be brought up into an operational mode.

- a. In the MI click **Stop** to stop the system software.
- b. Perform the Archive.
- c. In the MI click **Start** to start the system software. Wait for the system to complete “Diagnostic” mode and enter either Track or Maintenance mode.
- d. In the MI click **Test Mode** in the **System** menu.
- e. From the Test Mode dialog click **Command BIT** and wait for BIT to complete; repeat a minimum of five times.
- f. Exit Test Mode by clicking **Close**.
- g. Click **SYSTEM STATUS** to open the System Status window.
- h. Under **View**, select **Current** and wait for the screen to update.
- i. Verify that all of the LRU status indicators are green.
- j. If any of the status indicators are yellow or red, click on that button to see what alerts/alarms have been recorded for the LRU.
- k. Go back through the set up procedure for that particular nominal for the LRU.

4.21 System TOA Calibration

This procedure is designed to calibrate the system based on the Time of Arrival (TOA) data from the Cal/BIT signal. To do this the correct settings for monitor limits must have been entered into the system following the procedures in Section 4. Once those are entered, the SysCal tool is used to calculate the calibration offsets caused by to path delays from the cabling and the connection to each assembly. Data computed in the SysCal tool is entered into the MI Settings dialog upon completion of the procedure.

- a. In the MI click **System Calibration (SysCal)** in the **Tools** menu to open the SysCal tool [Figure 4-11].

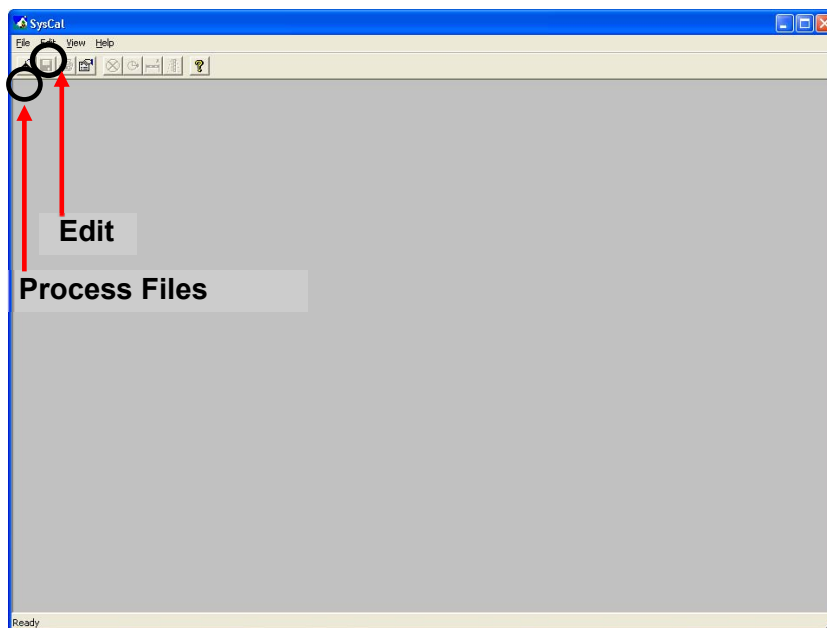


Figure 4-11 SysCal Tool

- b. In the SysCal Tool click **Settings** in the **Edit** menu. This will open the SysCal Settings dialog.
- c. Under **Processing Method** select **Calibrate from BIT data (use monitor_limits.dat)** [Figure 4-12].

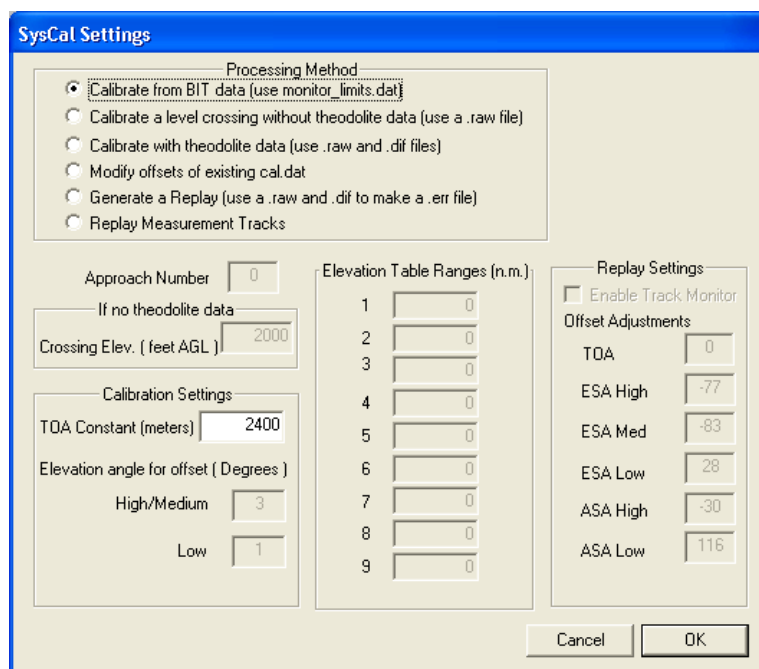


Figure 4-12 SysCal Settings Dialog

- d. Click **OK** to exit the SysCal Settings dialog
- e. In SysCal, click the **Process Files** icon [Figure 4-11]. This will open the Select monitor_limits.dat file dialog [Figure 4-13].

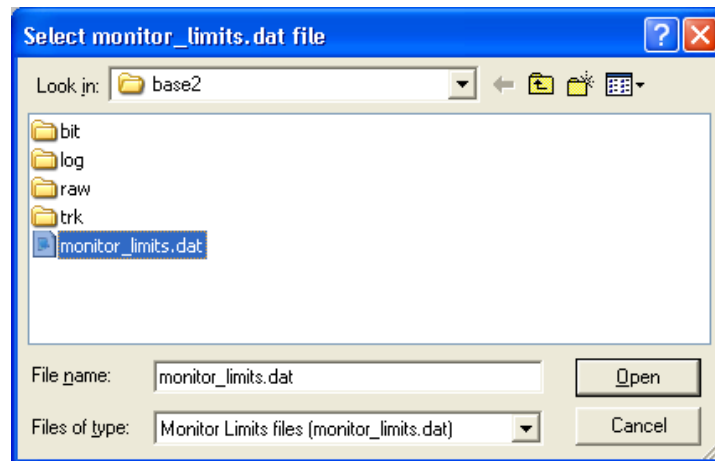


Figure 4-13 Select monitor_limits.dat dialog

- f. Select monitor_limits.dat and click **Open** button to process the file. This will bring up the Cal/BIT Antenna Z Coordinate dialog [Figure 4-14].

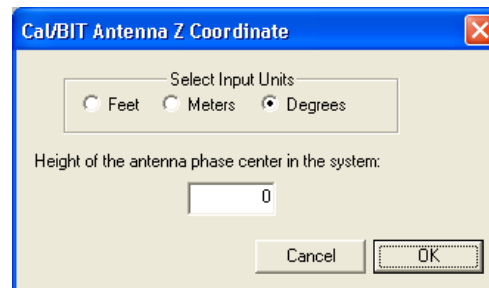
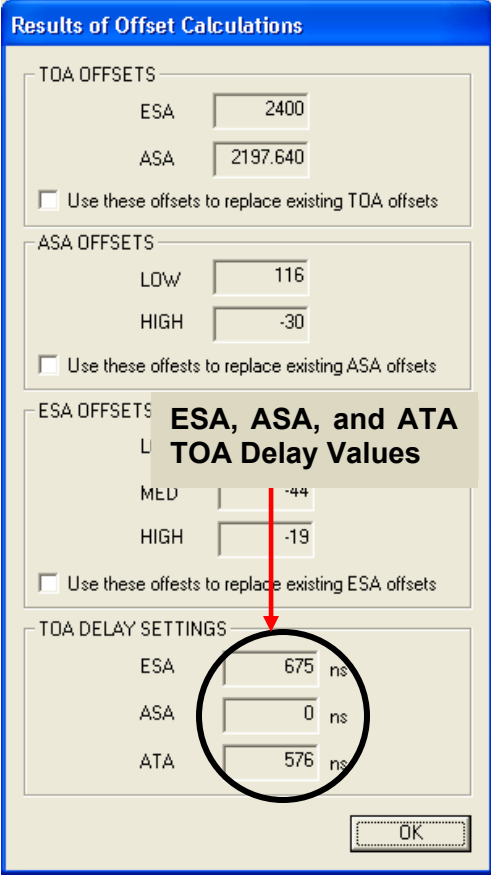


Figure 4-14 Cal/BIT Antenna Z Coordinate dialog

- g. In the Cal/BIT Antenna Z Coordinate dialog use the defaults and click **OK** to process the file. This will open the Results of Offset Calculations dialog [Figure 4-15].



The dialog box titled "Results of Offset Calculations" contains four sections: TOA OFFSETS, ASA OFFSETS, ESA OFFSETS, and TOA DELAY SETTINGS. The TOA DELAY SETTINGS section is circled, and a red arrow points to it from the text "ESA, ASA, and ATA TOA Delay Values" which is overlaid on the ESA OFFSETS section.

Section	Parameter	Value
TOA OFFSETS	ESA	2400
	ASA	2197.640
ASA OFFSETS	LOW	116
	HIGH	-30
ESA OFFSETS	MED	-44
	HIGH	-19
TOA DELAY SETTINGS	ESA	675 ns
	ASA	0 ns
	ATA	576 ns

Figure 4-15 Results of Offset Calculations dialog

- h. In the Results of Offset Calculations dialog locate the TOA DELAY SETTINGS. Record the three numbers for the ESA, ASA and ATA delays [Figure 4-15] then click **OK** to exit the dialog.
- i. In SysCal, click **Exit** in the **File** menu to exit the SysCal tool.
- j. In the MI click **SYSTEM SETUP**.
- k. In the System Setup dialog click **Settings**. This will open the Settings dialog [Figure 4-16].

System Settings

Diagnostic Mode BIT Schedule
 Retry Interval: 10 seconds
 Retries: 0
 Daily BIT Time: 12:00:00 AM

Maintenance Mode BIT Schedule
 Interval: 1 minutes

GTU Track BIT Schedule
 Interval: 60 minutes

F1 Cal/BIT Position
 Near: 5 μ s

UPS Shutdown
 10 minutes

Environment
 Barometric Pressure: 29.92 in Hg
 Transition Level: 18000 feet
 Max Noise Level: 40 counts
☐ Compress Sensor Data

GTU Type
☐ 920-00101 ☒ 920-00402
☒ Multi-GTU (920-00706) Installed
☒ GTU 1 Installed ☒ GTU 3 Installed
☐ GTU 2 Installed ☐ GTU 4 Installed

GCA Guidance
☒ Enable

Service Volume
 Az Volume Edges: +/- 35 degrees
 EI Volume Lower Edge: 0.6 degrees
 EI Volume Upper Edge: 7 degrees
 Localizer Range: 18 NM
 Glideslope R
 EI Az Volume Edges: +/- 10 degrees

Acquire Settings
 Acquire Timeout: 2 minutes
 Reacquire Limit: 5

Interrogation
 Guidance: 10 Hz
 Surveillance: 2 Hz
☐ Enable P2

TOA Delay Settings
 ESA: 0 ns ASA: 0 ns
 ATA: 0 ns

Data Recording
 Record Track RAM Files:
☒ Off ☐ Guidance ☐ All
 Record BIT RAM Files
 Record Raw Files
 Record GTU Files
☐ Record Track BIT Files
☒ Log Track Info Messages
☒ Log DQM Messages
 First Character of Raw File Name: x
 Send FTM Data for this RCU: RCU1

Rack Config
☒ Single ☐ Dual
 Rack A ☐ Rack B

OK Cancel Apply

Figure 4-16 MI Settings: Enter TOA Delay

- l. In the Settings dialog under **TOA Delay Settings**, enter the recorded values for ESA, ASA and ATA TOA delays [Figure 4-15].
- m. In the Settings dialog click **OK**.
- n. In the System Setup dialog click **Close**.

4.21.1 Pressure Sensor Verification

1. With the TLS software running, examine the field “Altimeter” at the lower-left corner of the RCU display and verify that the field displays altitude (should not show “N/A”). Record this verification.

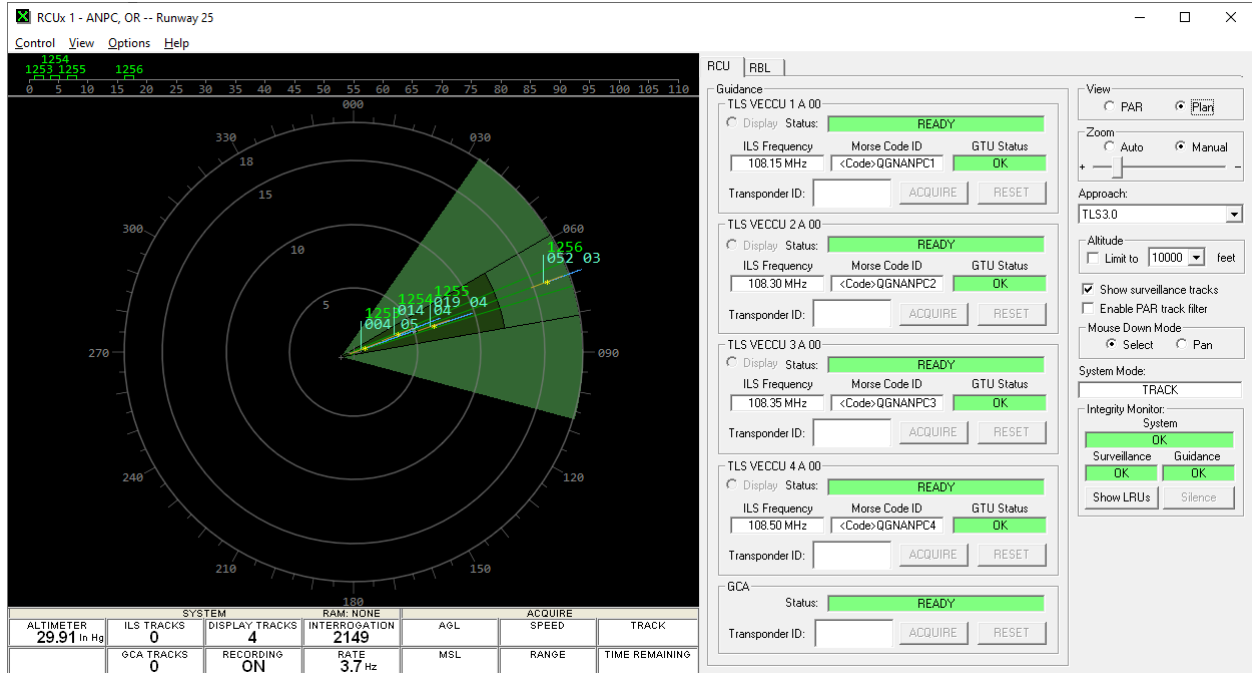


Figure 4-17 RCU Display of Altimeter

2. Verify that surveillance tracks appear on the RCU display and record this verification.

Appendix A Abbreviations and Definitions

Abbreviation	Definition
μs	Micro Seconds
ANPC	Advanced Navigation & Positioning Corporation
ASA	Azimuth Sensor Assembly (previously called LAOA)
ATA	Azimuth Time-Of-Arrival (TOA) Antenna
ATC	Air Traffic Control
BIT	Built-In-Test
Cal/BIT	Calibration/Built-in-Test
CNTL	Control
CPU	Central Processing Unit
CSC	Computer Software Component
dB	Decibel
DDM	Differential Depth of Modulation
DME	Distance Measurement Equipment
ESA	Elevation Sensor Assembly (previously called GAOA)
FAR	Federal Aviation Regulations
FTM	Flight Test Monitor
GS	Glide slope
GTU	Guidance Transmitter Unit
Hz	Hertz
ID	Identification
ILS	Instrument Landing System
IM	Integrity Monitor
KVM	Keyboard Video Mouse
LOC	Localizer
LRU	Line Replaceable Unit
nm	nanometer
NM	Nautical Mile
MΩ	Mega Ohms
MHz	Mega Hertz
MI	Maintenance Interface
MIU	Maintenance Interface Unit
OFZ	Obstacle Free Zone

Abbreviation	Definition
OPS	Operations
OS	Operating System
PM	Preventive Maintenance
RCU	Remote Control Unit
RF	Radio Frequency
RSA	Runway Safety Area
SCI	Software Configuration Item
SDM	Sum of Depth of Modulation
TLS	Transponder Landing System
TOA	Time-of-Arrival
TRR	Test Result Register
UPS	Uninterruptible Power Supply
VAC	Volts Alternating Current
VDC	Volts Direct Current
VHF	Very High Frequency

Appendix B ANPC TLS TI Manual Recommended Change Form

The following form is intended to document recommended changes pertaining to the Technical Instruction Manual. The TLS TI Manual Recommended Change Form may be legibly handwritten or typed and must include the revision level of the TI Manual being discussed.

The recommendation sheet is intended to be used as master copies from which to make working recommendation sheet.

Scan and email the completed form to Email CustomerSupport@anpc.com or use standard mail and post to this address:

ANPC
489 N. 8th Street, Suite 203
Hood River, OR 97031
ATTN: Customer Support

Phone: +1 (541) 386-1747

Appendix C Installation Checklist

Post this list on the wall of the shelter

- ☐ Coordinate date and time when calibration aircraft will be needed
- ☐ Install antenna components according to the siting plan
- ☐ Keep coax connects clean and dry; store dust caps for future use
- ☐ Pull fiber optic cable using only the strain relief, never by the connector
- ☐ Check cable labels during and after install to ensure cables are not swapped
- ☐ Tune the F1 delay to have jitter less than 2.5 ns for the ATA, ESA and ASA
- ☐ Measure survey distances in units of feet, enter survey distances in units of feet
- ☐ Carefully measure the ESA elevation, use the correct sign when inputting data
- ☐ Select the offset theodolite approach for calibration if applicable
- ☐ Collect the calibration data in test mode with the GTU's off
- ☐ Use the Mode-C calibration option if theodolite data is missing
- ☐ Set localizer volume to 25 nm and +/- 42 degrees
- ☐ Set glide slope volume to 12 nm +/- 12 degrees

TI MANUAL RECOMMENDED CHANGE FORM

This recommended change pertains to:

- ☐ Safety
- ☐ Operation
- ☐ Maintenance
- ☐ Documentation

Recommendation:

[illegible]